COMPANION WORKBOOK

PRACTICAL TRAINING PROFORMAS

Applied Basics of Rasashastra and Bhaishajya Kalpana

AYPG-AB-RB · MD / MS AYURVEDA · SEMESTER II

Printable proformas for all
47 Practical Training Activities of the NCISM
Competency-Based Dynamic Curriculum

Companion volume to the Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana

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PARUL INSTITUTE OF AYURVED AND RESEARCH
Vadodara · Gujarat · Bharat

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Applied Basics of Rasashastra and Bhaishajya Kalpana

AYPG-AB-RB · Semester II

Practical Training Proformas

Applied Basics of Rasashastra and Bhaishajya Kalpana (AYPG-AB-RB)

MD / MS Ayurveda, Semester II

First edition, 2026

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Companion volume

Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana, ISBN 978-81-955675-2-2.

Use of artificial intelligence

This workbook was created by Dr Milind Chatrabhuji using Claude Opus, an artificial intelligence assistant developed by Anthropic. Every page has been checked and approved by the author, who takes full responsibility for the content.

Safety notice

Several of the procedures recorded in this workbook use mercury, arsenic sulphides, lead and other toxic materials, strong acids, volatile solvents, high temperatures and laboratory animals. They are to be carried out only in a properly equipped laboratory, under the direct supervision of a qualified teacher, and in accordance with the rules in force. Neither the author nor the publisher accepts liability for any injury or loss arising from the use of this workbook.

Typeset in Roboto. Set to A4 for double-sided printing.

Printed in Bharat.

PREFACE

A practical is remembered by the hand, not by the eye. The scholar who has weighed the drug, watched the colour change and written the reading down at the moment it was taken will carry that knowledge into the examination hall and into practice. The scholar who copies a record a week later will not.

That is the reason for this workbook. The NCISM Competency-Based Dynamic Curriculum for Applied Basics of Rasashastra and Bhaishajya Kalpana lays down forty-seven Practical Training Activities across eight modules, and one hundred and sixty notional hours are given to them. Yet in most departments the scholar sits down with a blank journal and decides for himself what to record. Important observations are lost simply because no column was ruled for them.

Here every activity of Table 4 has been given a proforma of its own. The title is the curriculum's own. The aim, the allotment, the requirements, the procedure and the observation tables are already printed, drawn from the activity details that the curriculum lays down. What is left blank is only what the scholar must supply. The observation tables have been built around what the procedure actually produces, so that a weight, a temperature, a time or an Rf value has a place waiting for it.

Two things have been kept firmly in view. The first is safety. This subject handles mercury, arsenic sulphides and lead, it works at the temperature of a Puta, and it uses solvents, acids and living animals. Wherever the work is hazardous, a safety block is printed on the page before the procedure begins. It is not a formality. It is the part of the page I would ask every scholar to read first.

The second is honesty in recording. A proforma should not push the scholar towards the answer that is expected. Space has therefore been left for the result that does not match the classical description, and for the limitation that the method could not overcome. A procedure that failed and was recorded truthfully teaches more than one that succeeded on paper.

This workbook is a companion to the Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana. It is given free of cost, as that book was. If it saves a scholar from losing an observation, it has done its work.

Dr Milind Chatrabhuji
Parul Institute of Ayurved and Research

ACKNOWLEDGEMENTS

No workbook of this kind is the work of one person. It rests on the teachers who first demonstrated these procedures, and on the scholars whose questions showed where a column was missing.

I record my gratitude to the management and the administration of Parul University and to Parul Institute of Ayurved and Research, for agreeing to publish this work free of cost and for leaving the copyright with the author.

I thank the postgraduate scholars who used the draft proformas in their sessions and told me plainly where the writing space fell short. Much of the final layout is theirs.

I thank the National Commission for Indian System of Medicine, whose curriculum document is the foundation of this workbook.

Above all I thank my family, who gave up the evenings and the holidays that this work consumed.

Dr Milind Chatrabhuji

HOW TO USE THIS WORKBOOK

WHAT THIS WORKBOOK CONTAINS

One proforma for each of the forty-seven Practical Training Activities listed in Table 4 of the NCISM curriculum for AYPG-AB-RB. The activities appear in the order the curriculum gives them, module by module. The title of every proforma is the exact wording of the curriculum, in its own spelling, except that three misspellings are corrected: Synthesiis (8.4), Sharakara (7.7) and Pharmacopeial (7.6).

HOW EACH PROFORMA IS ARRANGED

Every proforma follows the same order, so that the scholar never has to look for a section. The module band at the top names the module. The line below it gives the practical number and the notional hours allotted to it. Then come the aim, the allotment, the requirements, the safety block where the work is hazardous, the procedure, the observation tables, the calculation, the result, the discussion, the conclusion, the precautions and the teacher's signature.

FOR THE SCHOLAR

1. Read the whole proforma before the session begins, and read the safety block first.
2. Fill the allotment as soon as the teacher gives it. An unrecorded allotment cannot be reconstructed later.
3. Write the observation at the bench, at the moment it is made. Do not trust memory and do not write it up afterwards.
4. Record a reading that does not agree with the expected value exactly as it was found. State the disagreement in the discussion. Do not adjust it.
5. Where a printed table row does not apply to your sample, write "not applicable" rather than leaving it blank.
6. Get the signature of the teacher before the next session.

FOR THE TEACHER

Allot a different drug, sample, database or instrument to each scholar wherever the curriculum asks for it, and avoid repeating an allotment in a later session. The blank rows left at the end of the printed tables are for the parameters you add. The proforma is a record, not an assessment sheet. Assess each module by the methods the curriculum gives at its end, and grade it as Table 6C directs.

PRINTING

The workbook is set to A4 with mirrored margins, so print it double-sided, flipping on the long edge. The inner margin is wider to allow for binding. Every writing box and every table stays whole on its page, so no answer is ever split across a leaf.

WHAT THIS WORKBOOK DOES NOT CONTAIN

It does not cover the forty-four Experiential Learning Activities of Table 5. Those are to be issued separately. It is not a textbook and it carries no descriptive matter; for that, use the Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana.

THE COURSE AT A GLANCE

AYPG-AB-RB is studied in Semester II. It carries 8 modules and 16 credits, which come to 480 notional learning hours. One credit is 30 hours of learner engagement, divided between teaching, practical training and experiential learning in the ratio of 1 : 2 : 3. So one credit gives 5 hours of teaching, 10 hours of practical training, 13 hours of experiential learning and 2 hours for the modular assessment.

MODULES, CREDITS AND HOURS

NO.	MODULE	CREDITS	LECTURE	PRACTICAL	EXPERIENTIAL	TOTAL	MARKS
M1	Pharmaceutics and Pharmaceutical Technology	2	10	20	30	60	50
M2	Databases and Information Systems for Metal, Mineral, and Plant Research	2	10	20	30	60	50
M3	Essential Chemistry - Unlocking the Rasashastra Techniques	2	10	20	30	60	50
M4	Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures	2	10	20	30	60	50
M5	Principles, specification, SOPs of handling of classical and advanced instruments	3	15	30	45	90	75
M6	Animal house and Animal handling technique	1	5	10	15	30	25
M7	Classical and Advanced quality control techniques for Ayurveda Formulations	2	10	20	30	60	50
M8	Nano-science, nano-technology and its application in Ayurveda	2	10	20	30	60	50
	Total	**16**	**80**	**160**	**240**	**480**	**400**

The experiential column includes the hours set aside for the modular assessment. The 160 practical training hours are the hours covered by this workbook.

MARKS FOR THE UNIVERSITY EXAMINATION

PAPER	THEORY	PRACTICAL	TOTAL
AYPG-AB-RB, Paper 1	100	200	300

THEORY PAPER PATTERN, 3 HOURS, 100 MARKS

TYPE OF QUESTION	NUMBER	MARKS EACH	TOTAL
Application-based question (ABQ)	1	20	20
Short answer questions (SAQ)	8	5	40
Analytical based structured long answer questions (LAQ)	4	10	40
Total			**100**

PRACTICAL EXAMINATION, 200 MARKS

NO.	HEAD	MARKS
1	Major practical - one pharmaceutical preparation of 60 to 90 minutes (40) and one analytical practical of 60 to 90 minutes (40)	80
2	Minor practical - one pharmaceutical preparation under 30 minutes (25) and one analytical practical under 30 minutes (25), together with spotting and identification of 10 samples (10)	60
3	Viva voce - general and structured	40
4	Practical record book carrying 25 entries	10
5	Log book	10
	Total	**200**

The 10 spotting samples are made up of 2 raw materials (metals, minerals, Visha Dravya), 2 finished products (one Rasoushadhi and one Kashthoushadhi), 2 manufacturing equipment, 2 analytical equipment and 2 analytical study reports. The curriculum spells Kashthoushadhi as "Kshthoushadhi"; the spelling is corrected here.

ABBREVIATIONS

The keys below are reproduced from the Abbreviations section of the NCISM curriculum document for AYPG-AB-RB, followed by the general abbreviations used in this workbook.

DOMAIN AND SUB-DOMAIN

CK	Cognitive / Knowledge	**PSY-MEC**	Psychomotor / Mechanism
CC	Cognitive / Comprehension	**PSY-ADT**	Psychomotor / Adaptation
CAP	Cognitive / Application	**PSY-ORG**	Psychomotor / Origination
CAN	Cognitive / Analysis	**AFT-REC**	Affective / Receiving
CS	Cognitive / Synthesis	**AFT-RES**	Affective / Responding
CE	Cognitive / Evaluation	**AFT-VAL**	Affective / Valuing
PSY-SET	Psychomotor / Set	**AFT-SET**	Affective / Organization
PSY-GUD	Psychomotor / Guided response	**AFT-CHR**	Affective / Characterization

LEVEL

K	Know	**SH**	Shows how
KH	Knows how	**D**	Does

TEACHING AND LEARNING METHODS

L	Lecture	**KL**	Kinaesthetic Learning
L&PPT	Lecture with PowerPoint presentation	**W**	Workshops
L&GD	Lecture and Group Discussion	**GBL**	Game-Based Learning
L_VC	Lecture with Video clips	**LS**	Library Session
REC	Recitation	**PL**	Peer Learning
SY	Symposium	**RLE**	Real-Life Experience
TUT	Tutorial	**PER**	Presentations
DIS	Discussions	**D-M**	Demonstration on Model
BS	Brainstorming	**PT**	Practical
IBL	Inquiry-Based Learning	**X-Ray**	X-ray Identification
PBL	Problem-Based Learning	**CD**	Case Diagnosis
CBL	Case-Based Learning	**LRI**	Lab Report Interpretation
PrBL	Project-Based Learning	**DA**	Drug Analysis
TBL	Team-Based Learning	**D**	Demonstration
TPW	Team Project Work	**D-BED**	Demonstration Bedside
FC	Flipped Classroom	**DL**	Demonstration Lab
BL	Blended Learning	**DG**	Demonstration Garden
EDU	Edutainment	**FV**	Field Visit
ML	Mobile Learning	**JC**	Journal Club
ECE	Early Clinical Exposure	**Mnt**	Mentoring
SIM	Simulation	**PAL**	Peer Assisted Learning
RP	Role Plays	**C_L**	Co Learning
SDL	Self-directed learning	**DSN**	Dissection
PSM	Problem-Solving Method	**PSN**	Prosection

GENERAL ABBREVIATIONS USED IN THIS WORKBOOK

AAS	Atomic Absorption Spectroscopy	**IR**	Infrared
ABQ	Application-Based Question	**ISBN**	International Standard Book Number
AFI	Ayurvedic Formulary of India	**LAQ**	Long Answer Question
AgNPs	Silver nanoparticles	**MeSH**	Medical Subject Headings
API	Ayurvedic Pharmacopoeia of India	**MGP**	Modular Grade Point
AYUSH	Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy	**NCISM**	National Commission for Indian System of Medicine
CCSEA	Committee for Control and Supervision of Experiments on Animals, formerly CPCSEA	**NMR**	Nuclear Magnetic Resonance
cGMP	current Good Manufacturing Practice	**PSA**	Particle Size Analyser
CO	Course Outcome	**Rf**	Retardation factor, in chromatography
CPCSEA	Committee for the Purpose of Control and Supervision of Experiments on Animals, now CCSEA	**SAQ**	Short Answer Question
DOI	Digital Object Identifier	**SEM**	Scanning Electron Microscopy
DTA	Differential Thermal Analysis	**SGPA**	Semester Grade Point Average
EDAX	Energy Dispersive X-ray Analysis	**SOP**	Standard Operating Procedure
FDA	Food and Drug Administration, United States	**SQL**	Structured Query Language
GLP	Good Laboratory Practice	**TEM**	Transmission Electron Microscopy
GMP	Good Manufacturing Practice	**TGA**	Thermogravimetric Analysis
HPTLC	High Performance Thin Layer Chromatography	**TLC**	Thin Layer Chromatography
IAEC	Institutional Animal Ethics Committee	**URL**	Uniform Resource Locator
ICP	Inductively Coupled Plasma	**USGS**	United States Geological Survey
ICP-MS	Inductively Coupled Plasma - Mass Spectrometry	**UV**	Ultraviolet
ICP-OES	Inductively Coupled Plasma - Optical Emission Spectrometry	**VSM**	Vibrating Sample Magnetometer
IMPPAT	Indian Medicinal Plants, Phytochemistry And Therapeutics	**XRD**	X-ray Diffraction
IPNI	International Plant Names Index	**XRF**	X-ray Fluorescence

CONTENTS

The 47 Practical Training Activities of AYPG-AB-RB, in the order given in Table 4 of the curriculum. Notional hours are shown against each.

Module 1 | Pharmaceutics and Pharmaceutical Technology | 8 practicals, 20 hours | page 1

1.1	Demonstration of regulatory frameworks (FDA, etc.)	2 h	**3**
1.2	Demonstration of pharmaceutical product development process	2 h	**8**
1.3	Identify and categorise Classical Ayurvedic dosage forms	2 h	**13**
1.4	Identification and categorisation of modified Ayurvedic dosage forms	2 h	**18**
1.5	Formulation of Solid Dosage Forms selecting appropriate excipients	4 h	**22**
1.6	Formulation of liquid Dosage Forms selecting appropriate excipients	4 h	**28**
1.7	GMP audit and compliance evaluation	2 h	**34**
1.8	Demonstration, designing, and evaluation of pharmaceutical packaging and labelling	2 h	**40**

Module 2 | Databases and Information Systems for Metal, Mineral, and Plant Research | 7 practicals, 20 hours | page 47

2.1	Demonstration of various searching techniques for databases	2 h	**49**
2.2	Demonstration of Modelling, Querying, and Analytics of Data	2 h	**54**
2.3	Demonstration of techniques to search and extract data from the USGS Mineral Database	2 h	**59**
2.4	Analyse and integrate Web mineral Database	3 h	**65**
2.5	Search and explore Metal Database and Applications	3 h	**71**
2.6	Demonstration of techniques to explore Plant Databases	4 h	**76**
2.7	Analysis of case studies of Database Applications in Metal, Mineral, or Plant Research	4 h	**81**

Module 3 | Essential Chemistry - Unlocking the Rasashastra Techniques | 5 practicals, 20 hours | page 87

3.1	Analysis of chemical bonding types, principles, and isomerism	5 h	**89**
3.2	Demonstration of the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures	5 h	**94**
3.3	Demonstration of different types of extraction methods including solvent selection	4 h	**100**
3.4	Demonstration of qualitative identification of phytoconstituents in single herbs and herbal formulations	3 h	**106**
3.5	Demonstration of different types of metals and their properties	3 h	**111**

Module 4 | Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures | 2 practicals, 20 hours | page 117

4.1	Demonstration of the Shodhana procedure of Rasadravyas using various methods	10 h	**119**
4.2	Demonstration of the Marana procedure of Rasadravyas with different methods	10 h	**126**

Module 5 | Principles, specification, SOPs of handling of classical and advanced instruments | 7 practicals, 30 hours | page 135

5.1	Demonstration of selected Puta to give the knowledge of Standard Operative Procedure	4 h	**137**

5.2	Demonstration of the handling of Musha following the Standard Operating Procedure (SOP)	4 h	**143**
5.3	Demonstration of the chromatography techniques to separate, identify components and interpret the results	6 h	**149**
5.4	Demonstration of spectroscopic instruments through Audiovisual aids	3 h	**156**
5.5	Demonstration of handling of petrological and trinocular microscopes	6 h	**161**
5.6	Demonstration of calibration of various instruments present in the department	3 h	**167**
5.7	Demonstration of characterization instruments with Audiovisual aids	4 h	**172**

Module 6 | Animal house and Animal handling technique | 5 practicals, 10 hours | page 177

6.1	Demonstration of animal house model	2 h	**179**
6.2	Demonstration of Animal Body Language	2 h	**185**
6.3	Demonstration of Species-Specific Handling Techniques for Laboratory Animals	2 h	**190**
6.4	Demonstration of microscopic Examination of Faecal Samples for Parasites	2 h	**195**
6.5	Evaluation of Animal Welfare and Ethics in a Research Setting	2 h	**201**

Module 7 | Classical and Advanced quality control techniques for Ayurveda Formulations | 8 practicals, 20 hours | page 207

7.1	Demonstration of documentation and record-keeping processes in the quality control section	2 h	**209**
7.2	Demonstration of assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)	2 h	**214**
7.3	Demonstration of Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva	3 h	**220**
7.4	Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti	3 h	**227**
7.5	Demonstration of quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana, Lauha/Mandura	4 h	**234**
7.6	Demonstrate Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopoeial standards	2 h	**241**
7.7	Demonstration of quality control parameters for liquid dosage forms like Kashaya, Arka, Sharkara, Asava-arishta	2 h	**248**
7.8	Demonstration of quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays	2 h	**255**

Module 8 | Nano-science, nano-technology and its application in Ayurveda | 5 practicals, 20 hours | page 261

8.1	Demonstration of the measurement of micro-particles using relevant equipment	4 h	**263**
8.2	Demonstration of particle size of Bhasma prepared with different numbers of Puta	4 h	**268**
8.3	Demonstration of particle size of various Kupipakwa Rasayana samples	4 h	**274**
8.4	Synthesis of silver nanoparticles (AgNPs) using the chemical precipitation method	4 h	**279**
8.5	Conversion of Taila into oil-in-water nanoemulsion and its characterization	4 h	**285**

RECORD OF COMPLETION

Fill one line as each activity is completed and signed. This sheet is the quickest way for the scholar, the teacher and the examiner to see what has been covered and what is still pending.

NO.	PRACTICAL TRAINING ACTIVITY	HRS	PAGE	DATE	TEACHER'S SIGNATURE
Module 1 - Pharmaceutics and Pharmaceutical Technology					
1.1	Demonstration of regulatory frameworks (FDA, etc.)	2	3		
1.2	Demonstration of pharmaceutical product development process	2	8		
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3.2	Demonstration of the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures	5	94		

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MODULE 1

Pharmaceutics and Pharmaceutical Technology

2 credits · 60 notional hours · 20 hours of practical training
8 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
1.1	Demonstration of regulatory frameworks (FDA, etc.)	2
1.2	Demonstration of pharmaceutical product development process	2
1.3	Identify and categorise Classical Ayurvedic dosage forms	2
1.4	Identification and categorisation of modified Ayurvedic dosage forms	2
1.5	Formulation of Solid Dosage Forms selecting appropriate excipients	4
1.6	Formulation of liquid Dosage Forms selecting appropriate excipients	4
1.7	GMP audit and compliance evaluation	2
1.8	Demonstration, designing, and evaluation of pharmaceutical packaging and labelling	2

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MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.1 — **2 hours**

Demonstration of regulatory frameworks (FDA, etc.)

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To follow the scenario of a company developing a new drug, to identify the regulations and the guidelines that govern the submission of an Investigational New Drug application, and to construct the process of that application.

ALLOTMENT

Drug candidate in the scenario		Disease condition	
Regulatory framework allotted		Country or region	
Resource or audio-visual aid used		Date of the session	

FDA stands for the Food and Drug Administration, the regulatory authority for drugs in the United States. The curriculum names it as an example; any comparable regulatory framework may be allotted.

REQUIREMENTS

Audio-visual aids, internet connection, and the reference documents and guidelines of the allotted regulatory authority.

PROCEDURE

The following steps were carried out.

1. The teacher set out the scenario of a company developing a new drug for a stated disease condition.
2. The purpose of pharmaceutical regulation was explained.
3. The structure of the allotted regulatory authority was shown.
4. The regulations and guidelines governing an Investigational New Drug application were identified.
5. The steps of the application were traced in order.
6. The Indian position under the Drugs and Cosmetics Act, 1940 and its Rules was placed beside it.
7. The student constructed the application process and presented it.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Profile of the regulatory framework

Name of the authority		Country or region	
Year of establishment			
Governing law			
Scope of regulation			
Official website			
Type of approval pathway		Time usually taken for approval	

Table 2. Steps in the Investigational New Drug application

NO.	STEP	DOCUMENT OR DATA REQUIRED	GUIDELINE THAT APPLIES
1			
2			
3			
4			
5			
6			
7			
8			

Table 3. Comparison with the Indian framework

POINT OF COMPARISON	ALLOTTED FRAMEWORK	INDIA
Governing law		
Regulatory authority		
Licensing requirement		
Quality standard followed		
Requirement of clinical data		
Labelling requirement		
Post-marketing surveillance		

THE APPLICATION PROCESS AS CONSTRUCTED AND PRESENTED

RELEVANCE TO AYURVEDIC PRODUCTS

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.2 — **2 hours**

Demonstration of pharmaceutical product development process

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the video demonstration of the pharmaceutical product development process and to record the work carried out at each stage, from the selection of the drug candidate to post-marketing surveillance.

ALLOTMENT

Drug candidate selected in the video		Dosage form	
Title and source of the video			
Duration of the video		Session or batch number	
Date of the session			

REQUIREMENTS

Audio-visual content of 15 to 25 minutes, projector or computer, and product literature.

PROCEDURE

The following steps were carried out.

1. The teacher introduced the selected drug candidate.
2. The video presentation was shown in full.
3. Each stage of development was noted as it appeared.
4. The documents generated at each stage were listed.
5. The time and the cost involved were discussed.
6. The common reasons for failure at each stage were noted.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Stages of product development

NO.	STAGE	WORK CARRIED OUT	DOCUMENT GENERATED	TIME FRAME
1	Introduction to the selected drug candidate			
2	Discovery and target identification			
3	Preclinical testing and formulation development			
4	Clinical trials			
5	Regulatory submission and approval			
6	Commercialisation and post-marketing surveillance			
7				
8				

Table 2. Profile of the selected drug candidate

Name of the product		Dosage form	
Composition			
Indication		Route of administration	
Dose		Shelf life declared	
Regulatory category			

CHALLENGES IDENTIFIED AND THE SOLUTIONS SUGGESTED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.3 — **2 hours**

Identify and categorise Classical Ayurvedic dosage forms

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To identify the classical Ayurvedic dosage forms placed before the student and to classify them into solid, liquid and semisolid categories.

ALLOTMENT

Number of samples given		Sample codes allotted	
Session or batch number		Date of the session	

REQUIREMENTS

Labelled samples of classical dosage forms, hand lens, spatula, watch glass, white tile, and the reference texts.

> **SAFETY** Some classical dosage forms contain mercury, arsenic or lead. Do not taste any sample unless the teacher directs it. Handle powders with a spatula, not with the fingers, and keep the samples away from the face. Wash the hands after the session.

PROCEDURE

The following steps were carried out.

1. Each sample was observed for its physical state and appearance.
2. The organoleptic characters were noted.
3. The dosage form was identified.
4. The sample was placed in the solid, liquid or semisolid category.
5. The Kalpana group to which it belongs was decided.
6. The classical reference for the form was traced.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Identification of the samples

NO.	SAMPLE CODE	DOSAGE FORM IDENTIFIED	PHYSICAL STATE	KALPANA GROUP	CHARACTERISTICS RECORDED
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Table 2. Categorisation of the samples

NO.	CATEGORY	DOSAGE FORMS PLACED UNDER IT
1	Solid dosage forms	
2	Liquid dosage forms	
3	Semisolid dosage forms	
4		
5		

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.4 — **2 hours**

Identification and categorisation of modified Ayurvedic dosage forms

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To identify the modified Ayurvedic dosage forms placed before the student and to classify them into solid, semisolid, liquid and gaseous categories.

ALLOTMENT

Number of samples given		Sample codes allotted	
Session or batch number		Date of the session	

Samples to be examined and sorted include tablets, capsules, solutions, suspensions, ointments, creams and infusions.

REQUIREMENTS

Labelled samples of modified dosage forms, their packs and labels, hand lens, and the reference texts.

> **SAFETY** Do not taste any sample, and do not open a sealed pack unless the teacher directs it. Point the nozzle of an inhaler or a spray away from the face. Return every sample to its labelled container. Wash the hands after the session.

PROCEDURE

The following steps were carried out.

1. Each sample and its pack were observed.
2. The modified dosage form was identified.
3. The physical state of the sample was decided.
4. The classical form from which it is derived was traced.
5. The purpose served by the modification was noted.
6. The samples were sorted into the four categories.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Identification of the samples

NO.	SAMPLE CODE	MODIFIED DOSAGE FORM	PHYSICAL STATE	CLASSICAL FORM IT IS DERIVED FROM	PURPOSE OF THE MODIFICATION
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Table 2. Categorisation by physical state

NO.	CATEGORY	DOSAGE FORMS PLACED UNDER IT
1	Solid dosage forms	
2	Semisolid dosage forms	
3	Liquid dosage forms	
4	Gaseous dosage forms	

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.5 — **4 hours**

Formulation of Solid Dosage Forms selecting appropriate excipients

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To formulate the allotted solid dosage form by selecting suitable excipients, and to record the process and the characters of the finished product.

ALLOTMENT

Dosage form allotted		Model drug allotted	
Batch size			
Reference formula followed			

The curriculum sets a simple tablet formulation using a model drug and excipients as the standard exercise. The ingredients are measured and mixed, the tablets are compressed, and the physical characters are evaluated.

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Wear a mask and gloves while handling fine powders. Keep the pulveriser and the tablet press guarded. Do not open any machine while it is running. Clean the equipment between batches to prevent cross contamination.

PROCEDURE

The following steps were carried out.

1. The formula was worked out for the given batch size.
2. Each ingredient was weighed accurately.
3. The powders were sifted through the required sieve.
4. The ingredients were mixed in the geometric order.
5. The tablets were compressed, or the chosen process was carried out.
6. The finished product was dried and weighed.
7. The product was evaluated for the required characters.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Formula worked out

NO.	INGREDIENT	FUNCTION SERVED	QUANTITY PER UNIT	QUANTITY PER BATCH
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 2. Process record

NO.	OPERATION	EQUIPMENT USED	TIME TAKEN	OBSERVATION
1				
2				
3				
4				
5				
6				
7				
8				

Table 3. Characters of the finished product

Appearance		Colour	
Odour		Taste	
Average weight		Weight variation	
Hardness		Friability	
Disintegration time		Yield obtained	
Any defect noticed			

CALCULATION

Yield (%) = Actual yield ÷ Theoretical yield × 100

Weight variation (%) = (Individual weight − Average weight) ÷ Average weight × 100

Friability (%) = (Initial weight − Final weight) ÷ Initial weight × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.6 — **4 hours**

Formulation of liquid Dosage Forms selecting appropriate excipients

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To formulate the allotted liquid dosage form by selecting suitable excipients, and to record the process and the characters of the finished product.

ALLOTMENT

Dosage form allotted		Model drug allotted	
Batch size			
Reference formula followed			

The curriculum sets a simple syrup formulation using a model drug and excipients as the standard exercise. The ingredients are measured and mixed, the solids are dissolved, and the physical and chemical characters are evaluated.

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Handle hot liquids with tongs and heat resistant gloves. Keep volatile solvents away from a flame. Label every container at once. Wash any spill with plenty of water.

PROCEDURE

The following steps were carried out.

1. The formula was worked out for the given batch size.
2. The vehicle and the excipients were selected and justified.
3. Each ingredient was weighed or measured accurately.
4. The solids were dissolved in the proper order.
5. The preparation was filtered or strained as required.
6. The volume was made up and the product was filled and labelled.
7. The product was evaluated for the required characters.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Formula worked out

NO.	INGREDIENT	FUNCTION SERVED	QUANTITY PER UNIT	QUANTITY PER BATCH
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 2. Process record

NO.	OPERATION	EQUIPMENT USED	TEMPERATURE (°C)	TIME TAKEN	OBSERVATION
1					
2					
3					
4					
5					
6					
7					
8					

Table 3. Characters of the finished product

Appearance		Colour	
Odour		Taste	
pH		Specific gravity	
Viscosity		Clarity	
Sedimentation observed		Redispersibility	
Volume obtained			
Any defect noticed			

CALCULATION

Yield (%) = Actual volume obtained ÷ Theoretical volume × 100

Specific gravity = Weight of the liquid ÷ Weight of an equal volume of water

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.7 — **2 hours**

GMP audit and compliance evaluation

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To conduct a mock audit of the allotted area using the checklist provided, and to evaluate compliance with Good Manufacturing Practice and current Good Manufacturing Practice.

AUDIT DETAILS

Unit or area audited			
Address of the unit			
Name of the auditor		Date of the audit	
Reference schedule followed		Licence number, if any	

Good Manufacturing Practice for Ayurvedic, Siddha and Unani medicines is laid down in Schedule T of the Drugs and Cosmetics Rules, 1945. Where the teaching pharmacy is not GMP certified, the mock audit is conducted all the same and the shortcomings and the components required to meet the standard are recorded.

REQUIREMENTS

Audit checklist provided by the teacher, measuring tape, camera where permitted, and a copy of the reference schedule.

> **SAFETY** Follow the entry rules of each area and wear the gown, cap, mask and footwear prescribed there. Do not touch a running machine, an open product or any material under process. Keep clear of furnaces, boilers and open fires. Stay with the group and leave an area at once if the supervisor asks.

PROCEDURE

The following steps were carried out.

1. The teacher explained the importance of GMP and cGMP in drug manufacturing.
2. The checklist was received and read before the audit began.
3. Each area was inspected in a fixed order.
4. Documentation, personnel training, equipment maintenance and facility cleanliness were assessed.
5. The observation was recorded against every point on the checklist.
6. The shortcomings were discussed and the corrective actions were suggested.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Audit checklist

NO.	REQUIREMENT CHECKED	COMPLIES	OBSERVATION	CORRECTIVE ACTION SUGGESTED
1	Location and surroundings			
2	Building and premises			
3	Water supply and its quality			
4	Disposal of waste			
5	Machinery and equipment			
6	Equipment maintenance records			
7	Personnel, health and hygiene			
8	Personnel training records			
9	Facility cleanliness and sanitation			
10	Raw material store			
11	Finished goods store			
12	Documentation and records			
13	Quality control section			
14				
15				

Table 2. Summary of the audit

Total points checked		Points complying	
Points not complying		Compliance (%)	
Major shortcoming noticed			
Components required to meet the standard			

CALCULATION

Compliance (%) = Number of points complying ÷ Total number of points checked × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 1	Pharmaceutics and Pharmaceutical Technology

PRACTICAL TRAINING 1.8 **2 hours**

Demonstration, designing, and evaluation of pharmaceutical packaging and labelling

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To design the packaging and the label for the allotted product, considering the container closure system, the labelling requirements and the tamper-evident features, and to evaluate the design against the regulatory requirements.

ALLOTMENT

Product allotted		Dosage form	
Pack size proposed		Container proposed	

REQUIREMENTS

Samples of packaging materials, labels of marketed products, measuring scale, drawing sheet, and the reference rules.

PROCEDURE

The following steps were carried out.

1. The teacher demonstrated a standard packaging and labelling design.
2. The properties of the product that decide the packing were listed.
3. A suitable container closure system was selected.
4. The tamper-evident feature was decided.
5. The label content was drafted.
6. The draft was checked against the regulatory requirements and industry standards.
7. The design was presented and evaluated.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Packaging design

PARTICULAR	PROPOSAL	JUSTIFICATION
Primary container		
Closure		
Container closure system as a whole		
Tamper-evident feature		
Secondary packing		
Material of construction		
Protection required		
Storage condition declared		
Surface available for the label		

Table 2. Label content checklist

NO.	PARTICULAR REQUIRED ON THE LABEL	PRESENT	REMARK
1	Name of the drug		
2	Complete list of ingredients with quantity		
3	Net content of the pack		
4	Batch number		
5	Date of manufacture		
6	Date of expiry or best use before		
7	Manufacturing licence number		
8	Name and full address of the manufacturer		
9	Dose, directions for use and Anupana		
10	Caution or warning where applicable		
11	Maximum retail price		
12			
13			

SPACE FOR THE LABEL DESIGN

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

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MODULE 2

Databases and Information Systems for Metal, Mineral, and Plant Research

2 credits · 60 notional hours · 20 hours of practical training
7 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
2.1	Demonstration of various searching techniques for databases	2
2.2	Demonstration of Modelling, Querying, and Analytics of Data	2
2.3	Demonstration of techniques to search and extract data from the USGS Mineral Database	2
2.4	Analyse and integrate Web mineral Database	3
2.5	Search and explore Metal Database and Applications	3
2.6	Demonstration of techniques to explore Plant Databases	4
2.7	Analysis of case studies of Database Applications in Metal, Mineral, or Plant Research	4

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MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.1 — **2 hours**

Demonstration of various searching techniques for databases

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe and practise the searching techniques that are commonly used in scientific databases, including the use of MeSH terms, and to compare the results that each technique returns.

ALLOTMENT

Database used			
Topic searched			
Session or batch number		Date of the session	

MeSH stands for Medical Subject Headings, the controlled vocabulary used to index articles in the biomedical databases.

REQUIREMENTS

Computer with an internet connection, a web browser, and access to the database.

PROCEDURE

The following steps were carried out.

1. The database was opened and its search page was located.
2. A simple keyword search was carried out first.
3. Each further technique was applied in turn.
4. The search was repeated using the appropriate MeSH terms.
5. The number of records returned by each search was noted.
6. The results were compared and the best search string was decided.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Database details

Name of the database		Hosting agency	
Full form			
URL used			
Type of access		Coverage of the database	
Date of access		Version or year	

Table 2. Search techniques practised

NO.	TECHNIQUE	SEARCH STRING USED	MESH OR INDEX TERM USED	RECORDS RETURNED	REMARK
1	Simple keyword search				
2	Phrase search within quotation marks				
3	Boolean AND				
4	Boolean OR				
5	Boolean NOT				
6	Truncation or wildcard				
7	Field specific search				
8	Search using MeSH terms				
9	Filters and limits applied				
10					
11					

COMPARISON OF THE RESULTS OBTAINED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.2 — **2 hours**

Demonstration of Modelling, Querying, and Analytics of Data

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe how a data model is created for a metal, mineral or plant database, to write structured queries against it, and to carry out the analytics tasks on the data retrieved.

ALLOTMENT

Dataset or database used			
Subject of the dataset			
Software used		Date of the session	

SQL stands for Structured Query Language.

REQUIREMENTS

Computer with an internet connection, database or spreadsheet software, and access to the dataset.

PROCEDURE

The following steps were carried out.

1. The teacher created a data model for the selected database.
2. The fields and the relations of the model were examined.
3. Structured queries were framed and run against the data.
4. The output of each query was recorded.
5. The analytics tasks were carried out on the data retrieved.
6. The output was summarised and interpreted.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Dataset details

Name of the dataset		Source	
URL used			
Number of records		Number of fields	
File format		Date of access	
Version or year			

Table 2. Data model

NO.	FIELD NAME	TYPE OF DATA	WHAT IT HOLDS	EXAMPLE VALUE
1				
2				
3				
4				
5				
6				
7				
8				

Table 3. SQL queries run and their output

NO.	QUERY IN PLAIN WORDS	SQL QUERY AS WRITTEN	RECORDS RETURNED	INFERENCE
1				
2				
3				
4				
5				
6				

Table 4. Data analytics tasks performed

NO.	TASK	METHOD OR TOOL USED	OUTPUT OBTAINED
1	Data visualisation		
2	Data mining		
3	Predictive modelling		
4			
5			

ANALYSIS OF THE OUTPUT

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.3 — **2 hours**

Demonstration of techniques to search and extract data from the USGS Mineral Database

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To search and extract data on the allotted mineral or metal from the USGS Mineral Database, and to analyse and present the data in a report.

ALLOTMENT

Mineral or metal allotted		Chemical name	
Session or batch number		Date of the session	

USGS stands for United States Geological Survey. The resources listed in the curriculum are the Minerals Yearbook for metals and minerals, and the USMIN Mineral Deposit Database. These give production, trade and deposit data. The physical properties of the mineral are recorded in Practical 2.4.

REQUIREMENTS

Computer with an internet connection, a web browser, and access to the USGS Mineral Database.

PROCEDURE

The following steps were carried out.

1. The USGS website was opened in the browser.
2. The minerals information section was located.
3. The allotted mineral or metal was searched by its name.
4. Suitable filters were applied to narrow the search.
5. The required data fields were noted and tabulated.
6. The data was analysed and presented in a report.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Database details

Name of the database		Hosting agency	
Full form			
URL used			
Country		Type of access	
Date of access		Version or year	

Table 2. Search record

NO.	SEARCH TERM USED	SECTION OR FILTER APPLIED	RECORDS FOUND	DATE OF ACCESS
1				
2				
3				
4				
5				
6				
7				

Table 3. Data extracted from the database

Name of the mineral or metal		Chemical formula	
Chemical composition			
Mode of occurrence and type of deposit			
Major producing countries			
Occurrence in India			
World mine production		World reserves	
Industrial uses			
Substitutes listed			
Price trend		Year of the data	

Table 4. Correlation with Rasashastra

PARTICULAR	ENTRY
Classical name	
Varga	
Reference text	
Classical description in brief	
Grahya and Agrahya Lakshana	
Shodhana media mentioned	

ANALYSIS AND INTERPRETATION FOR THE REPORT

LIMITATIONS OF THE DATABASE OBSERVED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.4 — **3 hours**

Analyse and integrate Web mineral Database

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To search the allotted mineral across several mineral and metal databases, to integrate the data so obtained in order to answer a real-world problem, and to record the challenges and the limitations of that integration.

ALLOTMENT

Mineral allotted		Chemical formula	
Real-world problem allotted			
Session or batch number			

REQUIREMENTS

Computer with an internet connection, a web browser, and the reference texts of Rasashastra.

PROCEDURE

The following steps were carried out.

1. Each database was opened in turn and the allotted mineral was searched.
2. The data available in each database was recorded separately.
3. The records were placed side by side and compared.
4. Any disagreement between the sources was noted.
5. The data was integrated to answer the allotted real-world problem.
6. The integrated record was presented and the limitations were discussed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Databases used

NO.	NAME OF THE DATABASE	URL USED	TYPE OF DATA OBTAINED	DATE OF ACCESS
1	Webmineral Mineral Database			
2	USGS Mineral Database			
3	Mineral database, as directed			
4	Metal database, as directed			
5				

Table 2. Data drawn from each database

PARTICULAR	DATABASE 1	DATABASE 2	DATABASE 3
Chemical formula			
Chemical composition			
Crystal system			
Hardness on the Mohs scale			
Specific gravity			
Optical property			
Mode of occurrence			
Production or reserve data			
Industrial uses			
Disagreement noticed, if any			

Table 3. Integration with the classical description

PARTICULAR	ENTRY FROM THE TEXT	REFERENCE
Classical name		
Varga		
Paryaya		
Grahya Lakshana		
Agrahya Lakshana		
Shodhana media		
Therapeutic use		

THE REAL-WORLD PROBLEM, AND HOW THE INTEGRATED DATA ANSWERS IT

CHALLENGES AND LIMITATIONS OF INTEGRATING THE DATABASES

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.5 — **3 hours**

Search and explore Metal Database and Applications

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To search a metal database for the allotted metal, to record its properties and its applications, and to present a case study on the use of metal databases in industry or in research.

ALLOTMENT

Metal allotted		Chemical symbol	
Case study selected			
Date of the session			

REQUIREMENTS

Computer with an internet connection, a web browser, and the reference texts.

PROCEDURE

The following steps were carried out.

1. The metal database was opened.
2. The allotted metal was searched.
3. The physical and the chemical properties were noted.
4. The ores and the method of extraction were noted.
5. The applications were listed field by field.
6. A case study on the use of metal databases was selected and prepared.
7. The case study was presented and discussed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Database details

Name of the database		Hosting agency	
Full form			
URL used			
Type of access		Date of access	

Table 2. Properties of the metal

Name of the metal		Chemical symbol	
Atomic number		Atomic weight	
Group and period		Density	
Melting point		Boiling point	
Common oxidation states			
Principal ores			
Method of extraction			
Classical name in Rasashastra			

Table 3. Applications recorded

NO.	FIELD OF USE	APPLICATION	SOURCE OF THE INFORMATION
1			
2			
3			
4			
5			
6			
7			

Table 4. Case study on the use of metal databases

Title of the case study	
Industry or research	
Source or reference	
Database applied and the purpose	
Outcome reported	
Value added by the database	

THE CASE STUDY AS PRESENTED AND DISCUSSED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.6 — **4 hours**

Demonstration of techniques to explore Plant Databases

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To search and extract data on the allotted plant from the plant databases, and to record its accepted botanical name, its synonyms and the phytochemical information available for it.

ALLOTMENT

Plant or family allotted		Sanskrit name	
Session or batch number		Date of the session	

The curriculum names The Plant List and the International Plant Names Index. The Plant List is no longer updated and has been superseded by World Flora Online, so the accepted name should be verified there. IPNI remains current and is used for the nomenclatural record.

REQUIREMENTS

Computer with an internet connection, a web browser, and the Nighantu texts.

PROCEDURE

The following steps were carried out.

1. Each database was opened in turn.
2. The allotted plant was searched by its botanical name.
3. The accepted name and the synonyms were recorded.
4. The nomenclatural details were checked on IPNI.
5. The phytochemical data available was gathered.
6. The Sanskrit name and the classical use were traced.
7. The data was analysed and the databases were compared.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Databases explored

NO.	NAME OF THE DATABASE	URL USED	TYPE OF INFORMATION OBTAINED	DATE OF ACCESS
1	The Plant List, as named in the curriculum			
2	World Flora Online, which supersedes it			
3	International Plant Names Index (IPNI)			
4	IMPPAT - Indian Medicinal Plants, Phytochemistry And Therapeutics			
5	Dr Duke's Phytochemical and Ethnobotanical Databases			
6	PubChem			
7				

Table 2. Profile of the plant

Accepted botanical name		Family	
Synonyms recorded			
Author citation from IPNI			
Sanskrit name		Part used	
Major phytoconstituents reported			
Reported pharmacological activity			
Source database used		Classical reference	
Rasa, Guna, Virya, Vipaka			

COMPARISON BETWEEN THE DATABASES EXPLORED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 2	Databases and Information Systems for Metal, Mineral, and Plant Research

PRACTICAL TRAINING 2.7 **4 hours**

Analysis of case studies of Database Applications in Metal, Mineral, or Plant Research

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To analyse the selected case studies of database applications, to evaluate how effective the application was, to identify the best practices, and to suggest improvements and future directions.

ALLOTMENT

Area allotted		Number of case studies analysed	
Session or batch number		Date of the session	

DOI stands for Digital Object Identifier, the permanent identifier printed on most journal articles. Record it wherever the article carries one.

REQUIREMENTS

Computer with an internet connection, access to peer-reviewed journals, and the printed copies of the selected case studies.

PROCEDURE

The following steps were carried out.

1. The case studies were searched and selected.
2. Only peer-reviewed articles were taken.
3. Each case study was read in full.
4. The database used and the purpose of its use were noted.
5. The effectiveness of the application was evaluated.
6. The best practices were identified.
7. Improvements and future directions were discussed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Case studies analysed

NO.	TITLE OF THE ARTICLE	AUTHOR AND YEAR	JOURNAL	DATABASE USED	DOI OR ACCESSION NUMBER
1					
2					
3					
4					

Table 2. Critical appraisal

POINT OF APPRAISAL	CASE STUDY 1	CASE STUDY 2	CASE STUDY 3
Objective of the study			
Database applied			
Method of data handling			
Main finding			
Effectiveness of the application			
Best practice identified			
Limitation of the study			
Improvement suggested			

FUTURE DIRECTIONS, AND APPLICATION TO YOUR OWN RESEARCH PLAN

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

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MODULE 3

Essential Chemistry - Unlocking the Rasashastra Techniques

2 credits · 60 notional hours · 20 hours of practical training
5 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
3.1	Analysis of chemical bonding types, principles, and isomerism	5
3.2	Demonstration of the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures	5
3.3	Demonstration of different types of extraction methods including solvent selection	4
3.4	Demonstration of qualitative identification of phytoconstituents in single herbs and herbal formulations	3
3.5	Demonstration of different types of metals and their properties	3

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MODULE 3	Essential Chemistry - Unlocking the Rasashastra Techniques

PRACTICAL TRAINING 3.1 **5 hours**

Analysis of chemical bonding types, principles, and isomerism

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To study the assigned video resources on chemical bonding and isomerism, to take notes on the concepts, and to present them with an interpretation relating to the Shodhana and the Marana process.

ALLOTMENT

Topic allotted			
Title and source of the video resource			
Duration of the resource		Date of the presentation	

REQUIREMENTS

Video resources assigned by the teacher, computer or projector, notebook, and the reference textbooks of chemistry.

PROCEDURE

The following steps were carried out.

1. The assigned video resources were watched in full.
2. Notes were taken on each type of chemical bonding.
3. Notes were taken on each type of isomerism.
4. An example was worked out for each type.
5. The concepts were related to the Shodhana and the Marana process.
6. The notes were presented and discussed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Types of chemical bonding

NO.	TYPE OF BOND	HOW IT IS FORMED	EXAMPLE	RELEVANCE TO SHODHANA OR MARANA
1	Ionic bond			
2	Covalent bond			
3	Coordinate bond			
4	Metallic bond			
5	Hydrogen bond			
6	Van der Waals forces			
7				

Table 2. Types of isomerism

NO.	TYPE OF ISOMERISM	BASIS OF THE DIFFERENCE	EXAMPLE
1	Chain isomerism		
2	Position isomerism		
3	Functional isomerism		
4	Metamerism		
5	Tautomerism		
6	Geometrical isomerism		
7	Optical isomerism		
8			

INTERPRETATION OF THESE CONCEPTS IN RELATION TO SHODHANA AND MARANA

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 3	Essential Chemistry - Unlocking the Rasashastra Techniques

PRACTICAL TRAINING 3.2 — **5 hours**

Demonstration of the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To attend the online lecture or webinar on the principles of crystallography and organometallic chemistry, and to present the key points and findings with reference to the procedures of Rasashastra.

DETAILS OF THE SESSION

Title of the online lecture or webinar			
Name of the resource person			
Organising department or institution		Platform used	
Date of the session		Duration of the session	

REQUIREMENTS

Computer with an internet connection, projector, notebook, and the reference textbooks.

PROCEDURE

The following steps were carried out.

1. The department arranged the online lecture or webinar.
2. The meaning of a crystal and a unit cell was explained.
3. The seven crystal systems and their relations were presented.
4. The applications of crystallography in understanding compound structures were discussed.
5. The principles of organometallic chemistry were taken up.
6. The students noted the key points and presented their findings.
7. Ideas were exchanged in the peer discussion that followed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. The crystal systems

NO.	CRYSTAL SYSTEM	AXIAL RELATION	ANGULAR RELATION	MINERAL EXAMPLE
1	Cubic			
2	Tetragonal			
3	Orthorhombic			
4	Hexagonal			
5	Trigonal			
6	Monoclinic			
7	Triclinic			

Table 2. Principles of crystallography covered in the session

NO.	PRINCIPLE	EXPLANATION IN BRIEF	SIGNIFICANCE IN UNDERSTANDING COMPOUND STRUCTURE
1			
2			
3			
4			
5			
6			
7			
8			

Table 3. Organometallic principles covered in the session

NO.	PRINCIPLE	EXPLANATION IN BRIEF	BEARING ON RASASHASTRA PROCEDURES
1			
2			
3			
4			
5			
6			

KEY POINTS AND FINDINGS PRESENTED BY THE STUDENT

POINTS RAISED IN THE PEER DISCUSSION

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 3	Essential Chemistry - Unlocking the Rasashastra Techniques

PRACTICAL TRAINING 3.3 — **4 hours**

Demonstration of different types of extraction methods including solvent selection

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the extraction of the allotted herb and of a formulation by different methods and solvents, and to record the percentage yield and the physical status of each extract.

ALLOTMENT

Herb allotted		Part of the plant used	
Formulation allotted		Dosage form of the formulation	
Methods demonstrated			
Solvents used			

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Alcohol, ether and petroleum ether catch fire easily. Keep them away from any flame and use a water bath for heating. Chloroform does not burn, but its vapour is harmful to breathe. Work in a well ventilated room or a fume hood. Wear gloves and eye protection. Never pipette any solvent by mouth.

PROCEDURE

The following steps were carried out.

1. The herb and the formulation were identified, dried and powdered.
2. A known weight of each was taken for every extraction.
3. The solvent was selected for each method and the choice was justified.
4. Each method of extraction was carried out in turn.
5. Every extract was filtered and concentrated.
6. Each dish was dried to a constant weight and weighed.
7. The percentage yield and the physical status of each extract were recorded.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Methods of extraction demonstrated

NO.	METHOD	PRINCIPLE IN BRIEF	SOLVENT COMMONLY USED	SUITABLE FOR
1	Maceration			
2	Percolation			
3	Digestion			
4	Decoction			
5	Infusion			
6	Soxhlet extraction			
7				

Table 2. Extraction record

NO.	SAMPLE	METHOD	SOLVENT	WEIGHT TAKEN (gm)	WEIGHT OF EXTRACT (gm)	YIELD (%)	PHYSICAL STATUS OF THE EXTRACT
1							
2							
3							
4							
5							
6							
7							
8							

CALCULATION

Yield of the extract (%) = Weight of the dried extract ÷ Weight of the drug taken × 100

BASIS ON WHICH EACH SOLVENT WAS SELECTED

COMPARISON BETWEEN THE HERB AND THE FORMULATION

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 3	Essential Chemistry - Unlocking the Rasashastra Techniques

PRACTICAL TRAINING 3.4 **3 hours**

Demonstration of qualitative identification of phytoconstituents in single herbs and herbal formulations

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To extract the allotted herb and formulation in the specified solvents and to test the extracts for phytoconstituents by test tube methods.

ALLOTMENT

Herb allotted		Part used	
Formulation allotted		Dosage form of the formulation	
Solvents used for extraction			
Date of the session			

REQUIREMENTS

NO.	APPARATUS, REAGENT OR CHEMICAL	GRADE OR SPECIFICATION	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Concentrated sulphuric acid and hydrochloric acid cause severe burns. Always add acid to water and never the reverse. Use a fume hood for the Salkowski and the Molisch tests. Wear gloves and eye protection. Keep a wash bottle and running water within reach.

PROCEDURE

The following steps were carried out.

1. The herb and the formulation were extracted in the specified solvents.
2. Each extract was filtered and labelled.
3. A small quantity of the extract was taken in a clean test tube for each test.
4. The reagent was added as directed.
5. The change observed was recorded at once.
6. The inference was drawn against the standard observation.
7. A blank was run wherever the colour was doubtful.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Qualitative test tube methods carried out

NO.	GROUP TESTED FOR	NAME OF THE TEST	REAGENT USED	OBSERVATION IN THE HERB	OBSERVATION IN THE FORMULATION
1	Alkaloids	Dragendorff test			
2	Alkaloids	Mayer test			
3	Cardiac glycosides	Keller-Killiani test			
4	Flavonoids	Shinoda test			
5	Tannins	Ferric chloride test			
6	Tannins	Lead acetate test			
7	Saponins	Foam test			
8	Steroids and triterpenoids	Salkowski test			
9	Proteins	Biuret test			
10	Carbohydrates	Molisch test			
11	Fixed oils and fats	Spot test			
12					
13					

Table 2. Summary of the findings

PARTICULAR	HERB	FORMULATION
Groups found present		
Groups found absent		
Group present in the largest amount		
Doubtful result, if any		
Overall inference		

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 3	Essential Chemistry - Unlocking the Rasashastra Techniques

PRACTICAL TRAINING 3.5 **3 hours**

Demonstration of different types of metals and their properties

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To identify the metal samples placed before the student and to record their properties, including malleability, ductility, heat conductivity, electrical conductivity, density and reactivity with acids.

ALLOTMENT

Number of samples given		Sample codes allotted	
Session or batch number		Date of the session	

REQUIREMENTS

NO.	APPARATUS, REAGENT OR SAMPLE	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Dilute acids are corrosive. Add the acid to the metal in a small quantity and behind a shield. Some metals give off hydrogen, which is inflammable. Hold hot metal with tongs only. Lead and its compounds are toxic. Wash the hands after handling every sample.

PROCEDURE

The following steps were carried out.

1. Each sample was observed for its colour, lustre and form.
2. Each property was tested in turn on every sample.
3. The density was determined by weight and displacement.
4. The heat and the electrical conductivity were tested.
5. The reactivity of the sample with a dilute acid was noted.
6. The metal was identified from the properties recorded.
7. The classical name and the Varga of the metal were traced.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Identification of the samples

NO.	SAMPLE CODE	METAL IDENTIFIED	CLASSICAL NAME	COLOUR AND LUSTRE	BASIS OF IDENTIFICATION
1					
2					
3					
4					
5					
6					
7					
8					

Table 2. Properties recorded

NO.	PROPERTY	METHOD OF TESTING	OBSERVATION
1	Malleability		
2	Ductility		
3	Heat conductivity		
4	Electrical conductivity		
5	Density		
6	Reactivity with acids		
7	Hardness		
8	Lustre		
9	Sonority		
10	Behaviour on heating		
11			

CALCULATION

Density = Mass of the sample ÷ Volume of the sample

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 4

Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

2 credits · 60 notional hours · 20 hours of practical training
2 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
4.1	Demonstration of the Shodhana procedure of Rasadravyas using various methods	10
4.2	Demonstration of the Marana procedure of Rasadravyas with different methods	10

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MODULE 4	Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

PRACTICAL TRAINING 4.1 — **10 hours**

Demonstration of the Shodhana procedure of Rasadravyas using various methods

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe and record the Standard Operating Procedure of the Shodhana of the allotted Rasadravya by the allotted method, and to understand the rationale behind the Shodhana Dravya used.

ALLOTMENT

Rasadravya allotted		Varga or group	
Shodhana method allotted		Shodhana Dravya used	
Reference text followed			
Session or batch number			

Methods of Shodhana covered under this practical: Nimajjana (immersion), Bhavana (levigation), Swedana (heating under a liquid bath), Pachana (boiling), Patana (sublimation or distillation), Nirvapa (quenching), Dhalana (melting and pouring), Bharjana (roasting or frying) and Nirjaleekarana (dehydration).

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Parada, Haratala, Manahshila and Naga are toxic, and the fumes of molten Naga and Vanga must not be breathed. Work only in a well ventilated area or under an exhaust hood. Wear gloves, an apron, a mask and eye protection throughout. Never handle Parada with bare hands and never heat it in an open vessel. Collect every spill of Parada at once with a dropper or with Gandhaka, never with a broom. In Dhalana and Nirvapa the molten or red-hot metal spatters violently when it meets the liquid. Wear a face shield, pour from a low height through a perforated lid, and keep everyone else well back. Keep food and drink out of the laboratory. Collect all washings and residues for disposal as hazardous waste. Wash the hands and the face before leaving the laboratory.

CLASSICAL REFERENCE

TEXT	CHAPTER	VERSE NO.	STATEMENT IN BRIEF

PROCEDURE

The following steps were carried out.

1. The raw drug was identified and its organoleptic characters were recorded.
2. The initial weight of the raw drug was taken.
3. The Shodhana Dravya was prepared as directed in the reference text.
4. The allotted Shodhana method was carried out step by step.
5. The temperature and the duration were recorded for every step.
6. Washing and drying were carried out as directed.
7. The final weight of the Shodhita drug was recorded.
8. The Shodhita drug was examined for the features described in the text.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Raw material before Shodhana

Name of the drug		Source or supplier	
Batch or lot number		Quantity taken	
Colour		Odour	
Touch		Impurity noticed, if any	

Table 2. Process record

NO.	SHODHANA DRAVYA USED	QUANTITY OF THE MEDIA	TEMPERATURE (°C)	DURATION	OBSERVATION
1					
2					
3					
4					
5					
6					
7					
8					

Table 3. Weight record

STAGE	WEIGHT (gm)	LOSS OR GAIN (gm)	PERCENTAGE
Initial weight of the raw drug			
After cycle 1			
After cycle 2			
After cycle 3			
After cycle 4			
After the final washing			
After drying			
Final weight of the Shodhita drug			

Table 4. Comparison before and after Shodhana

PARAMETER	BEFORE SHODHANA	AFTER SHODHANA
Colour		
Odour		
Touch		
Lustre		
Brittleness		
Any other change		

CALCULATION

Percentage of loss = (Initial weight − Final weight) ÷ Initial weight × 100

SHODHITA LAKSHANA DESCRIBED IN THE TEXT AND OBSERVED

RATIONALE BEHIND THE SHODHANA DRAVYA USED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 4	Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

PRACTICAL TRAINING 4.2 **10 hours**

Demonstration of the Marana procedure of Rasadravyas with different methods

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe and record the Standard Operating Procedure of three Marana procedures carried out on the allotted Rasadravya, and to confirm the completion of each by the classical tests.

ALLOTMENT

Shodhita Rasadravya allotted		Varga or group	
Marana method 1 allotted		Marana method 2 allotted	
Marana method 3 allotted		Type of Puta used	
Bhavana Dravya used			
Reference text followed			

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Marana is carried out at a high temperature and often on toxic materials. Confirm that the drug has been properly Shodhita before starting. Wear gloves, an apron, a mask and eye protection. Keep the Sharava Samputa sealed so that no fume escapes. Do not stand over the pit while it burns. Allow Swangasheeta, self cooling, before opening. Never quench the Samputa with water. Grind the Bhasma in a closed area or under an exhaust and never dry sweep the spilled powder. Taste a Bhasma for Gatarasatva only when the teacher directs it, and only after it has passed the other tests. Collect all residues for disposal as hazardous waste.

CLASSICAL REFERENCE

TEXT	CHAPTER	VERSE NO.	STATEMENT IN BRIEF

PROCEDURE

The following steps were carried out.

1. The Shodhita drug was weighed and its characters were recorded.
2. The three Marana methods to be demonstrated were set out with their SOPs.
3. Kajjali or the mixture was prepared where the method requires it.
4. Bhavana was given with the allotted media and the mixture was dried.
5. Chakrikas were prepared, dried and sealed in the Sharava Samputa.
6. The allotted Puta was given and the temperature was recorded.
7. Swangasheeta was allowed and the Samputa was opened.
8. Each product was weighed, ground and tested for the Siddhi Lakshana.

Procedure as actually carried out:

OBSERVATIONS

Table 1. The three Marana methods demonstrated

NO.	METHOD	SOP STEPS IN BRIEF	MARANA DRAVYA USED	RATIONALE FOR ITS USE
1				
2				
3				

Table 2. Puta record

METHOD	PUTA NO.	BHAVANA DRAVYA	QUANTITY OF MEDIA	TYPE OF PUTA	MAXIMUM TEMPERATURE (°C)	COLOUR AFTER THE PUTA

Table 3. Weight record

STAGE	METHOD 1 (gm)	METHOD 2 (gm)	METHOD 3 (gm)
Weight of the Shodhita drug			
After Bhavana			
After Puta 1			
After Puta 3			
After Puta 5			
After the final Puta			
Final weight of the Bhasma			

Table 4. Bhasma Pariksha of each product

NO.	TEST	METHOD 1	METHOD 2	METHOD 3	INFERENCE
1	Nischandratva				
2	Rekhapurnatva				
3	Varitaratva				
4	Unama				
5	Slakshnatva				
6	Nirdhumatva				
7	Apunarbhava				
8	Niruttha				
9	Gatarasatva				
10	Avami				
11					

CALCULATION

Percentage of yield = Final weight of the Bhasma ÷ Weight of the Shodhita drug taken × 100

Percentage of loss = (Initial weight − Final weight) ÷ Initial weight × 100

COMPARISON BETWEEN THE THREE METHODS

BHASMA SIDDHI LAKSHANA DESCRIBED IN THE TEXT AND OBSERVED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

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MODULE 5

Principles, specification, SOPs of handling of classical and advanced instruments

3 credits · 90 notional hours · 30 hours of practical training
7 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
5.1	Demonstration of selected Puta to give the knowledge of Standard Operative Procedure	4
5.2	Demonstration of the handling of Musha following the Standard Operating Procedure (SOP)	4
5.3	Demonstration of the chromatography techniques to separate, identify components and interpret the results	6
5.4	Demonstration of spectroscopic instruments through Audiovisual aids	3
5.5	Demonstration of handling of petrological and trinocular microscopes	6
5.6	Demonstration of calibration of various instruments present in the department	3
5.7	Demonstration of characterization instruments with Audiovisual aids	4

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MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.1 **4 hours**

Demonstration of selected Puta to give the knowledge of Standard Operative Procedure

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the Standard Operating Procedure of the selected Puta with the selected fuel, and to record and document the temperature pattern and the time frame.

ALLOTMENT

Puta selected		Type of fuel selected	
Quantity of the fuel		Sample subjected to the Puta	
Reference text followed			
Session or batch number			

A different Puta and a different fuel are selected for every batch, so that the same combination is not repeated.

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY A Puta reaches several hundred degrees. Keep the pit away from any building and from stored fuel. Wear heat resistant gloves, an apron, covered shoes and eye protection. Stand upwind of the smoke. Keep sand and water at hand for an emergency. Do not disturb or quench the burning pit. Allow Swangasheeta, self cooling, before opening the Samputa.

PROCEDURE

The following steps were carried out.

1. The pit was measured and prepared as described in the text.
2. The fuel was arranged and weighed.
3. The Sharava Samputa was sealed, dried and placed in position.
4. The pit was ignited and the time was noted.
5. The temperature was recorded at fixed intervals.
6. The pit was allowed to cool by itself.
7. The Samputa was opened and the product was examined.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Specification of the Puta

Name of the Puta			
Dimensions of the pit			
Type of fuel		Quantity of the fuel	
Number of cow dung cakes		Position of the Sharava Samputa	
Method of ignition		Expected temperature	
Reference text and verse			

Table 2. Temperature and time record

NO.	TIME FROM IGNITION	TEMPERATURE (°C)	STAGE OF BURNING	OBSERVATION
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 3. Standard Operating Procedure drawn up

NO.	STEP	CRITICAL CONTROL POINT	PRECAUTION
1			
2			
3			
4			
5			
6			
7			
8			

SWANGASHEETA AND THE CONDITION OF THE PRODUCT ON OPENING

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.2 — **4 hours**

Demonstration of the handling of Musha following the Standard Operating Procedure (SOP)

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the use of a Musha or crucible for determining the ash value of the selected drug, and to record the temperature required, the time frame and the confirmatory test.

ALLOTMENT

Drug selected		Type of Musha used	
Quantity of the drug taken			
Reference text followed			

A different drug is selected for every batch, so that the same drug is not repeated.

REQUIREMENTS

NO.	APPARATUS, EQUIPMENT OR MATERIAL	SPECIFICATION OR SOURCE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY A Musha is handled red hot. Use long tongs and heat resistant gloves and keep the face away from the mouth of the furnace. A wet or incompletely dried Musha can crack and throw hot material. Dry it fully before use. Keep the area clear of any inflammable material.

PROCEDURE

The following steps were carried out.

1. The Musha was prepared, shaped and dried, and checked for cracks.
2. The empty Musha was weighed.
3. A known weight of the drug was placed in the Musha.
4. The Musha was heated in the furnace at the required temperature.
5. The temperature and the duration were recorded.
6. The Musha was cooled in a desiccator and weighed.
7. Heating was repeated to a constant weight.
8. The confirmatory test was applied to the ash obtained.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Specification of the Musha

Type of Musha		Shape and size	
Material used and proportion			
Method of preparation		Drying time given	
Capacity		Purpose of use	
Reference text and verse			

Table 2. Standard Operating Procedure drawn up

NO.	STEP	CRITICAL CONTROL POINT	PRECAUTION
1			
2			
3			
4			
5			
6			
7			
8			

Table 3. Determination of the ash value

PARTICULAR	OBSERVATION
Weight of the empty Musha	
Weight of the drug taken	
Temperature maintained	
Duration of heating	
Weight of the Musha with the ash	
Weight of the ash obtained	
Ash value (%)	
Colour and nature of the ash	
Confirmatory test applied	
Condition of the Musha after use	

CALCULATION

Ash value (%) = Weight of the ash obtained ÷ Weight of the drug taken × 100

Loss on ignition (%) = (Weight of the drug taken − Weight of the ash) ÷ Weight of the drug taken × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.3 — **6 hours**

Demonstration of the chromatography techniques to separate, identify components and interpret the results

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the chromatography of the selected herb and of a formulation containing that herb, to calculate the Rf values of the separated spots, and to interpret the result.

ALLOTMENT

Technique used		Herb selected	
Formulation selected, containing that herb			
Stationary phase used			
Mobile phase and ratio			
Detection method used			

The formulation may be a Kwatha, Churna, Sneha, Avaleha, Asava-Arishta or a tablet containing the selected herb as one of its ingredients. HPTLC of the same drug and formulation is carried out wherever the facility is available.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Methanol, toluene and ethyl acetate catch fire easily - keep them away from any flame. Chloroform does not burn, but its vapour is harmful, and formic acid is corrosive. Prepare and run the mobile phase in a fume hood. Keep the chamber closed. Use an ultraviolet cabinet with the shield down, since direct ultraviolet light injures the eyes and the skin. Wear gloves and eye protection. Collect the used solvent in a labelled waste bottle.

PROCEDURE

The following steps were carried out.

1. The herb and the formulation were extracted and the solutions were filtered.
2. The plate was activated and the origin line was marked.
3. Both samples and the standard were applied at the origin.
4. The chamber was saturated with the mobile phase.
5. The plate was developed to the marked front.
6. The plate was dried and the spots were detected.
7. The distances were measured and the Rf values were calculated.
8. The two chromatograms were compared.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Chromatographic conditions

Technique used		Stationary phase	
Mobile phase and ratio			
Method of application		Volume applied	
Chamber saturation time		Distance allowed to run	
Detection method		Temperature	
Humidity			

Table 2. Rf values

SPOT NO.	SAMPLE	DISTANCE MOVED BY THE SPOT (cm)	DISTANCE MOVED BY THE SOLVENT FRONT (cm)	RF VALUE	COLOUR IN VISIBLE LIGHT	COLOUR UNDER UV 254 nm	COLOUR UNDER UV 366 nm
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

CALCULATION

Rf = Distance moved by the solute ÷ Distance moved by the solvent front

SPACE FOR THE CHROMATOGRAM

INTERPRETATION, AND COMPARISON OF THE HERB WITH THE FORMULATION

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.4 — **3 hours**

Demonstration of spectroscopic instruments through Audiovisual aids

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To study the instrumentation and the working of the common spectroscopic instruments through audio-visual content, and to record the key points observed and the specification of one instrument in detail.

ALLOTMENT

Instrument allotted for detailed study			
Audio-visual resource used			
Session or batch number		Date of the session	

The working of these instruments is to be observed again during the visit to the characterisation laboratory under Module 8.

REQUIREMENTS

Audio-visual content, projector, and the instrument manuals where available.

PROCEDURE

The following steps were carried out.

1. The audio-visual content was shown instrument by instrument.
2. The principle of each instrument was noted.
3. The main parts of each instrument were identified.
4. The sample requirement of each was noted.
5. The students discussed the key points they had observed.
6. The teacher summarised the contents.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Instruments studied

NO.	INSTRUMENT	PRINCIPLE IN BRIEF	SAMPLE REQUIREMENT	INFORMATION OBTAINED
1	UV-Visible spectrophotometer			
2	Infrared spectrophotometer			
3	Nuclear Magnetic Resonance spectrometer			
4	Atomic Absorption Spectroscopy			
5	Inductively Coupled Plasma - Optical Emission Spectrometry or Mass Spectrometry (ICP-OES, ICP-MS)			
6	Flame photometer			
7				

Table 2. Specification of the instrument studied in detail

Name of the instrument		Make and model	
Source of radiation		Detector used	
Wavelength range		Sample holder used	
Software used		Resolution	
Application in Rasashastra			
Common fault and its remedy			

KEY POINTS OBSERVED AND DISCUSSED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.5 **6 hours**

Demonstration of handling of petrological and trinocular microscopes

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To handle the petrological and the trinocular microscope, to identify the selected mineral by its optical characters, and to record the morphology and the particle size of the sample.

ALLOTMENT

Mineral or sample selected		Form of the sample	
Microscope used		Magnification used	

A different sample is selected for every batch, so that the same sample is not repeated.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR MATERIAL	SPECIFICATION	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Handle the objective lenses only by the turret. Never force the coarse adjustment when a high power objective is in place. Switch off the lamp before changing the bulb and allow it to cool. Clean the lenses only with lens paper. Keep the slide preparation area free of any spilled mounting medium.

PROCEDURE

The following steps were carried out.

1. The parts of the microscope were identified and their functions were noted.
2. The illumination was adjusted and the field was centred.
3. The slide of the selected sample was prepared.
4. The sample was observed under plane polarised light.
5. The analyser was inserted and the sample was observed under crossed nicols.
6. The mineral was identified from its optical characters.
7. The morphology and the particle size were recorded with the help of projection, and the findings were discussed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Parts of the microscope and their functions

NO.	PART	FUNCTION
1	Eyepiece	
2	Objective lenses	
3	Polariser	
4	Analyser	
5	Rotating stage	
6	Bertrand lens	
7	Condenser	
8	Light source	
9	Camera port	
10	Ocular micrometer	

Table 2. Optical characters and morphology of the sample

Name of the sample		Form of the sample	
Method of slide preparation			
Magnification used		Colour in plane polarised light	
Pleochroism		Relief	
Birefringence		Extinction angle	
Shape of the particles			
Approximate particle size			
Mineral identified			

SPACE FOR THE DRAWING OF THE FIELD OBSERVED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.6 — **3 hours**

Demonstration of calibration of various instruments present in the department

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the methods of calibration of the instruments present in the department, as demonstrated by the concerned expert, and to record the deviation found in each.

ALLOTMENT

Instrument allotted for detailed record		Make and model	
Name of the expert who demonstrated			
Date of the session			

REQUIREMENTS

Calibration standards and certified reference weights, thermometer, standard buffer solutions, calibration register, and the instrument manuals.

SAFETY Switch off and unplug an instrument before it is opened or cleaned. Let hot air ovens, muffle furnaces and water baths cool before they are handled. Lift certified weights with forceps only. Wear gloves for buffer solutions and reagents. Do not change an instrument setting unless the expert directs it.

PROCEDURE

The following steps were carried out.

1. The calibration certificate and the due date of each instrument were checked.
2. The standard used for each instrument was identified.
3. The expert demonstrated the calibration of each instrument.
4. The observed value was compared with the standard value.
5. The deviation was calculated and recorded.
6. The entry was made in the calibration register and the next due date was fixed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Calibration record

NO.	INSTRUMENT	PARAMETER CALIBRATED	STANDARD USED	STANDARD VALUE	OBSERVED VALUE	DEVIATION	NEXT DUE DATE
1							
2							
3							
4							
5							
6							
7							
8							

Table 2. Method of calibration of the allotted instrument

NO.	STEP	ACCEPTANCE CRITERION
1		
2		
3		
4		
5		
6		
7		
8		

CALCULATION

Deviation (%) = (Observed value − Standard value) ÷ Standard value × 100

ACTION TAKEN WHEN THE DEVIATION IS BEYOND THE ACCEPTANCE LIMIT

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 5	Principles, specification, SOPs of handling of classical and advanced instruments

PRACTICAL TRAINING 5.7 — **4 hours**

Demonstration of characterization instruments with Audiovisual aids

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To search and collect articles on the instruments used for the characterisation of Ayurvedic formulations, to study their working through audio-visual aids, and to present the findings.

ALLOTMENT

Instrument allotted		Number of articles collected	
Session or batch number		Date of the presentation	

Only articles published in peer-reviewed journals are to be selected. The articles collected must be unique and different for each student, and the same article will not be accepted in a later session.

REQUIREMENTS

Computer with an internet connection, access to peer-reviewed journals, and printed copies of the selected articles.

PROCEDURE

The following steps were carried out.

1. The articles were searched in peer-reviewed journals.
2. The articles relating to the allotted instrument were selected.
3. Each article was read and its method and findings were noted.
4. The working of the instrument was studied through the audio-visual aids.
5. The working principle of the instrument was written down.
6. The findings were presented in the department and discussed.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Articles collected

NO.	INSTRUMENT	TITLE OF THE ARTICLE	JOURNAL AND YEAR	DOI OR ACCESSION NUMBER
1	PSA			
2	XRD			
3	XRF			
4	SEM-EDAX			
5	TEM			
6	TGA			
7	DTA			
8	VSM			

Table 2. Profile of the allotted instrument

PARTICULAR	ENTRY
Full form of the instrument	
Working principle	
Form of the sample required	
Quantity of the sample required	
Information obtained	
Unit of the result	
Limitation of the technique	
Significance in the standardisation of Ayurvedic medicines	

FINDINGS PRESENTED AND DISCUSSED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 6

Animal house and Animal handling technique

1 credits · 30 notional hours · 10 hours of practical training
5 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
6.1	Demonstration of animal house model	2
6.2	Demonstration of Animal Body Language	2
6.3	Demonstration of Species-Specific Handling Techniques for Laboratory Animals	2
6.4	Demonstration of microscopic Examination of Faecal Samples for Parasites	2
6.5	Evaluation of Animal Welfare and Ethics in a Research Setting	2

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MODULE 6	Animal house and Animal handling technique

PRACTICAL TRAINING 6.1 — **2 hours**

Demonstration of animal house model

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To study the animal house model demonstrated by the teacher, to describe each of its compartments, and to record the requirements of space, ventilation, temperature control, lighting, noise, sanitation, hygiene and record-keeping.

DETAILS OF THE SESSION

Type of model demonstrated			
Species the model is designed for		Scale of the model	
Session or batch number		Name of the teacher	
Date of the session			

In India, a facility that breeds or uses animals for research is registered with the CCSEA, the Committee for Control and Supervision of Experiments on Animals. The body was earlier called CPCSEA and was renamed in January 2023. Its requirements govern the design and the record-keeping described here.

REQUIREMENTS

Animal house model, measuring scale, observation checklist, notebook, and the reference guidelines.

> **SAFETY** Where the model is kept in or beside a live animal area, wear the clothing and footwear provided, do not touch or feed any animal, and follow the one-way movement from the clean area to the dirty area. Animal dander and bedding dust cause allergy. Wash the hands before leaving.

PROCEDURE

The following steps were carried out.

1. The teacher demonstrated the animal house model.
2. Each compartment of the model was identified and its purpose noted.
3. The space requirement of each compartment was recorded.
4. Ventilation, temperature control and lighting were described.
5. Noise levels, sanitation and hygiene practices were described.
6. The record-keeping and documentation system was explained.
7. The layout of the model was drawn.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Compartments of the model

NO.	COMPARTMENT	PURPOSE	SPACE REQUIREMENT	OBSERVATION
1	Quarantine room			
2	Animal holding room			
3	Experiment room			
4	Surgery room			
5	Store room			
6	Washing and sterilisation area			
7	Waste disposal area			
8	Record room			
9				

Table 2. Environmental requirements described

Space per animal		Ventilation and air changes	
Temperature control		Relative humidity	
Lighting and the light-dark cycle		Noise level permitted	
Sanitation practice		Hygiene practice	
Bedding material		Type of caging	

Table 3. Record-keeping and documentation

NO.	RECORD TO BE MAINTAINED	PURPOSE	FREQUENCY OF ENTRY
1	Animal stock and inventory register		
2	Breeding record		
3	Health monitoring record		
4	Feed and water record		
5	Environmental monitoring record		
6	Experiment protocol record		
7	Mortality and euthanasia record		
8			

SPACE FOR THE LAYOUT DRAWING OF THE ANIMAL HOUSE MODEL

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 6	Animal house and Animal handling technique

PRACTICAL TRAINING 6.2 **2 hours**

Demonstration of Animal Body Language

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the allotted group of laboratory animals for thirty minutes using the checklist of behavioural patterns and body language cues, and to interpret what each observed cue indicates.

ALLOTMENT

Species and group allotted		Number of animals observed	
Time of starting the observation		Duration of the observation	
Place of observation		Date of the session	

The observation period set by the curriculum is thirty minutes. A virtual observation using computer assisted learning software may be carried out before the live observation.

REQUIREMENTS

Observation checklist, stopwatch, notebook, and a camera where permitted.

> **SAFETY** Observe quietly and from a distance. A startled animal may bite or scratch. Do not put the hand into a cage during observation. Report any bite or scratch at once.

PROCEDURE

The following steps were carried out.

1. The checklist of behavioural patterns and body language cues was received.
2. The animals were observed without disturbing them for thirty minutes.
3. Every cue seen was entered on the checklist as it occurred.
4. The time and the duration of each behaviour were noted.
5. The findings were discussed and interpreted.
6. The findings were presented to the class.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Checklist of behavioural and body language cues

NO.	CATEGORY OF CUE	CUE OBSERVED	NUMBER OF TIMES	INTERPRETATION
1	Posture			
2	Facial expression			
3	Vocalisation			
4	Movement and gait			
5	Grooming			
6	Feeding behaviour			
7	Social interaction with the group			
8	Response to approach			
9				
10				

Table 2. Signs of distress and of comfort

NO.	SIGN OBSERVED	DISTRESS OR COMFORT	ACTION REQUIRED
1			
2			
3			
4			
5			
6			
7			
8			

IMPORTANCE OF RECOGNISING ANIMAL BEHAVIOUR AND BODY LANGUAGE

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 6	Animal house and Animal handling technique

PRACTICAL TRAINING 6.3 **2 hours**

Demonstration of Species-Specific Handling Techniques for Laboratory Animals

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To practise, under the guidance of the instructor, the handling and restraint techniques proper to each species of small laboratory animal, with attention to behaviour, body language and welfare.

ALLOTMENT

Species allotted		Number of animals handled	
Method of restraint allotted		Name of the supervisor	

REQUIREMENTS

Gloves, restrainers, towels, transfer cages, and the animals provided by the facility.

> **SAFETY** Handle an animal only under supervision. Wear gloves. Never lift a rat or a mouse by the tip of the tail. Support the body of a rabbit at all times, since its spine fractures easily. Return the animal to its cage gently. Report every bite or scratch and take the prescribed first aid at once.

PROCEDURE

The following steps were carried out.

1. The proper method of restraint for the species was demonstrated.
2. The site of holding was shown.
3. The student handled the animal under direct supervision.
4. The behaviour and the body language of the animal were watched throughout.
5. The welfare of the animal was considered at every step.
6. The animal was returned to its cage and observed again.
7. The session closed with a debriefing discussion.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Handling techniques for each species

NO.	SPECIES	METHOD OF RESTRAINT	SITE OF HOLDING	PRECAUTION
1	Mouse			
2	Rat			
3	Guinea pig			
4	Rabbit			
5				
6				

Table 2. Practice record

Species handled		Number of animals	
Duration of handling		Method of restraint used	
Difficulty faced			
Sign of distress noticed		Injury to the animal, if any	
Injury to the handler, if any			

DEBRIEFING - CHALLENGES, BEST PRACTICES AND FUTURE APPLICATIONS

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 6	Animal house and Animal handling technique

PRACTICAL TRAINING 6.4 **2 hours**

Demonstration of microscopic Examination of Faecal Samples for Parasites

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To collect and label faecal samples from the laboratory animals, to prepare stained slides, to examine them under the microscope at 40x and then at 100x, and to record the presence, the morphology and the size of any parasite seen.

ALLOTMENT

Species from which the samples were taken		Number of samples examined	
Staining solution used		Date of the session	

REQUIREMENTS

NO.	APPARATUS, REAGENT OR MATERIAL	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Faecal material may carry organisms that infect a human being. Wear gloves and a laboratory coat throughout and never eat or drink in the laboratory. Use a disposable applicator for each sample. Place all used slides, gloves and applicators in the biohazard bin. Disinfect the bench before and after the work. Wash the hands thoroughly.

PROCEDURE

The following steps were carried out.

1. The samples were collected fresh and labelled, wearing gloves and a coat.
2. The consistency and the colour of each sample were noted.
3. The smear was prepared and the staining solution was applied.
4. The slide was examined first at 40x magnification.
5. The magnification was increased to 100x for any structure seen.
6. The presence, the morphology and the size of the structure were recorded.
7. The findings were compared with the reference material.
8. The session closed with a debriefing discussion.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Methods used

NO.	METHOD	PRINCIPLE IN BRIEF	REAGENT OR STAIN USED
1	Direct smear		
2	Saline wet mount		
3	Iodine wet mount		
4	Stained smear		
5	Floatation method		
6	Sedimentation method		

Table 2. Observation

NO.	SAMPLE CODE	CONSISTENCY	MAGNIFICATION	STRUCTURE SEEN	MORPHOLOGY	SIZE (µm)	INFERENCE
1							
2							
3							
4							
5							
6							

Table 3. Comparison with the reference material

NO.	STRUCTURE SEEN	REFERENCE MATERIAL CONSULTED	PARASITE IDENTIFIED	DEGREE OF MATCH
1				
2				
3				
4				
5				

SPACE FOR THE DRAWING OF THE STRUCTURES SEEN

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 6	Animal house and Animal handling technique

PRACTICAL TRAINING 6.5 **2 hours**

Evaluation of Animal Welfare and Ethics in a Research Setting

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To evaluate the allotted case study scenario for its ethical and animal welfare issues using the animal welfare and ethics checklist, and to state the concerns and the recommendations that follow.

ALLOTMENT

Title of the case study scenario			
Species involved		Number of animals in the scenario	
Source of the scenario			
Session or batch number		Date of the session	

Animal experimentation in India is supervised by the CCSEA, formerly the CPCSEA, renamed in January 2023. Approval for a study is granted by the Institutional Animal Ethics Committee, the IAEC. The checklist below is applied to the scenario, not to a live facility.

REQUIREMENTS

Case study scenarios prepared by the teacher, the animal welfare and ethics checklist, and the reference guidelines.

PROCEDURE

The following steps were carried out.

1. The case study scenario was read in full.
2. The purpose, the species and the procedures described were noted.
3. Each point of the welfare and ethics checklist was applied to the scenario.
4. The ethical and welfare issues were identified.
5. The concerns were set down with the principle each one offends.
6. Recommendations were framed for each concern.
7. The findings were presented and discussed with the teacher.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Summary of the case study scenario

Title of the scenario			
Species involved		Number of animals	
Purpose of the study			
Procedure described			
Duration of the study			
Outcome reported			
Source of the scenario			

Table 2. Animal welfare and ethics checklist applied to the scenario

NO.	POINT EVALUATED	COMPLIES	OBSERVATION FROM THE SCENARIO
1	Registration of the facility with the regulatory authority		
2	Approval of the Institutional Animal Ethics Committee		
3	Approved protocol available for the study		
4	The three Rs applied - replacement, reduction, refinement		
5	Justification of the number of animals used		
6	Housing, caging and environmental enrichment		
7	Quality of the feed and the water		
8	Health monitoring during the study		
9	Policy for analgesia and anaesthesia		
10	Humane endpoints defined		
11	Approved method of euthanasia		
12			
13			

Table 3. Concerns and recommendations

NO.	CONCERN IDENTIFIED	PRINCIPLE IT OFFENDS	RECOMMENDATION
1			
2			
3			
4			
5			
6			

Table 4. Summary of the evaluation

Points evaluated		Points complying	
Points not complying		Points not stated in the scenario	
Major ethical concern			
Principal recommendation			

ETHICAL ISSUES DISCUSSED WITH THE TEACHER

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7

Classical and Advanced quality control techniques for Ayurveda Formulations

2 credits · 60 notional hours · 20 hours of practical training
8 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
7.1	Demonstration of documentation and record-keeping processes in the quality control section	2
7.2	Demonstration of assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)	2
7.3	Demonstration of Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva	3
7.4	Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti	3
7.5	Demonstration of quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana, Lauha/Mandura	4
7.6	Demonstrate Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopoeial standards	2
7.7	Demonstration of quality control parameters for liquid dosage forms like Kashaya, Arka, Sharkara, Asava-arishta	2
7.8	Demonstration of quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays	2

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MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.1 — **2 hours**

Demonstration of documentation and record-keeping processes in the quality control section

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To study, in the quality control section, the documentation and record-keeping processes followed for the different dosage forms, from the input materials to the finished products.

DETAILS

Name of the unit visited			
Address of the unit			
Name of the person in charge		Date of the visit	
Dosage forms covered			
Session or batch number			

REQUIREMENTS

Permission letter, specimen copies of the records, and a notebook.

> **SAFETY** The quality control section handles acids, solvents and hot equipment. Wear a laboratory coat, do not touch any reagent, sample or instrument in use, and do not eat or drink in the section.

PROCEDURE

The following steps were carried out.

1. Permission to study the records was obtained.
2. The teacher took the students to the quality control section.
3. Each record maintained in the section was listed.
4. The purpose of each record was noted.
5. The flow of an input material through to the finished product was traced.
6. One certificate of analysis was studied in full.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Records maintained in the quality control section

NO.	RECORD	PURPOSE	RETENTION PERIOD	PERSON RESPONSIBLE
1	Raw material inward register			
2	Sample receipt register			
3	Analytical worksheet			
4	Instrument log book			
5	Calibration record			
6	Standard operating procedures			
7	Certificate of analysis			
8	Stability study record			
9	Out of specification record			
10	Reference standard register			
11				

Table 2. Flow of documentation, from input material to finished product

NO.	STAGE	DOCUMENT RAISED	CHECKED BY	APPROVED BY
1	Receipt of the raw material			
2	Sampling of the raw material			
3	Testing of the raw material			
4	In-process checks			
5	Testing of the finished product			
6	Release of the batch			
7				

Table 3. Certificate of analysis studied

Name of the product		Batch number	
Date of manufacture		Date of analysis	
Quantity of the batch		Name of the analyst	
Approved by		Retest or expiry date	
Summary of the result			
Reference specification used			

OBSERVATIONS ON THE DOCUMENTATION SYSTEM

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.2 — **2 hours**

Demonstration of assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To apply the classical Bhasma Pariksha, together with the special tests for Rasadravyas, to one Bhasma from each category, to compare each with a negative sample, and to discuss the scientific basis of each test.

SESSION DETAILS

Number of samples tested			
Reference text followed			
Session or batch number		Date of the session	

B1 to B3 in the tables below stand for the three Bhasma listed in Table 1, and N for the negative sample. B1 to B3 are expected to be properly prepared, and so serve as the positive samples. N is a Bhasma that has not attained Siddhi. It is tested alongside them, so that each test shows both a positive and a negative result.

REQUIREMENTS

NO.	APPARATUS, REAGENT OR MATERIAL	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Several Bhasma are prepared from toxic metals. Do not taste any sample unless the teacher directs it and the sample is certified. Wear a mask while handling the fine powder, since it becomes airborne easily. The Nirdhumatva and the Apunarbhava tests use an open flame - keep inflammable material away and use tongs. In Amla Pariksha the Bhasma is kept with a sour medium such as curd or lemon juice; do not taste the mixture. Never taste the negative sample N; write 'not performed' against Gatarasatva and Avami for it. Wash the hands after the work.

PROCEDURE

The following steps were carried out.

1. One Bhasma was selected from each category - ore or mineral, metal, and Sudha Varga.
2. A negative sample, one that had not attained Siddhi, was kept ready.
3. Each sample was spread on a clean white tile and observed.
4. Each classical test was carried out in the described manner on all four samples, B1 to B3 and N.
5. The special tests for Rasadravyas were performed where they apply.
6. The observation was recorded immediately after each test.
7. The scientific basis of each test was discussed.
8. The overall conclusion on the Siddhi was arrived at for every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples tested

CODE	CATEGORY	NAME OF THE BHASMA	PREPARED FROM	NUMBER OF PUTA GIVEN	REFERENCE TEXT
B1	Ore or mineral origin				
B2	Metal origin				
B3	Sudha Varga				
N	Negative sample, Siddhi not attained				

Table 2. Classical Bhasma Pariksha

NO.	TEST	METHOD IN BRIEF	B1	B2	B3	N	INFERENCE
1	Nischandratva						
2	Rekhapurnatva						
3	Varitaratva						
4	Unama						
5	Slakshnatva						
6	Nirdhumatva						
7	Apunarbhava						
8	Niruttha						
9	Gatarasatva						
10	Avami						
11							

Table 3. Special Bhasma Pariksha for Rasadravyas

NO.	TEST	APPLIES TO	SAMPLE TESTED	METHOD IN BRIEF	OBSERVATION	INFERENCE
1	Amla Pariksha	Copper-containing compounds				
2						
3						
4						

Table 4. Result for each sample

CODE	CATEGORY	TESTS PASSED	TESTS FAILED	OVERALL INFERENCE
B1	Ore or mineral origin			
B2	Metal origin			
B3	Sudha Varga			
N	Negative sample			

SCIENTIFIC BASIS OF THE TESTS, AS DISCUSSED

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.3

3 hours

Demonstration of Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the quality control testing of one powder dosage form selected from each category named above, as per the parameters prescribed in the Pharmacopoeia, and to state whether each sample complies with the limits.

SESSION DETAILS

Number of samples tested			
Reference pharmacopoeia used			
Session or batch number		Date of the session	

S1 to S6 in the tables below stand for the samples listed in Table 1, in that order.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Muffle furnace and hot air oven surfaces stay hot for a long time. Use tongs and heat resistant gloves. Handle acids and solvents in a fume hood and add acid to water, never the reverse. Wear an apron and eye protection. Label every solution. Dispose of the residues of heavy metal testing as hazardous waste. Taste a sample only when the teacher directs it, and never taste an Anjana, an eye or ear preparation, or a Kshara meant for external use.

PROCEDURE

The following steps were carried out.

1. One formulation was selected from each category named in the title.
2. Each sample was received, labelled and its details were recorded.
3. The organoleptic characters of every sample were examined.
4. Each sample was weighed accurately for each test.
5. Every test was performed by the pharmacopoeial method.
6. The readings were recorded and the calculations were made.
7. The observed values were compared with the limits.
8. The inference was drawn for every parameter of every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples selected, one from each category

NO.	CATEGORY	NAME OF THE FORMULATION	MANUFACTURER	BATCH NUMBER	REFERENCE MONOGRAPH
1	Churna				
2	Kwatha Churna				
3	Lepa Churna				
4	Kshara				
5	Lavana				
6	Satva				

Table 2. Organoleptic characters of every sample

SAMPLE	COLOUR	ODOUR	TASTE	TOUCH	APPEARANCE
Churna					
Kwatha Churna					
Lepa Churna					
Kshara					
Lavana					
Satva					

Table 3. Physicochemical parameters

NO.	TEST	PHARMACOPOEIAL LIMIT	S1	S2	S3	S4	S5	S6
1	Loss on drying at 105 °C							
2	Total ash							
3	Acid insoluble ash							
4	Alcohol soluble extractive							
5	Water soluble extractive							
6	pH, at the concentration given in the monograph							
7	Fineness by sieve number							
8	Foreign matter							
9								
10								
11								

Table 4. Category specific tests, as directed by the teacher

NO.	SAMPLE	TEST	METHOD	OBSERVATION	INFERENCE
1					
2					
3					
4					
5					

Table 5. Result against the pharmacopoeial limits

SAMPLE	CATEGORY	PARAMETER NOT COMPLYING, IF ANY	COMPLIES / DOES NOT COMPLY
S1	Churna		
S2	Kwatha Churna		
S3	Lepa Churna		
S4	Kshara		
S5	Lavana		
S6	Satva		

CALCULATION

Loss on drying (%) = (Initial weight − Final weight) ÷ Initial weight × 100

Total ash (%) = Weight of the ash ÷ Weight of the sample taken × 100

Acid insoluble ash (%) = Weight of the acid insoluble ash ÷ Weight of the sample taken × 100

Extractive value (%) = Weight of the dried extract ÷ Weight of the sample taken × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.4 — **3 hours**

Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the quality control testing of one solid dosage form selected from each category named above, as per the parameters prescribed in the Pharmacopoeia, and to state whether each sample complies with the limits.

SESSION DETAILS

Number of samples tested			
Reference pharmacopoeia used			
Session or batch number		Date of the session	

S1 to S3 in the tables below stand for the samples listed in Table 1, in that order.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Muffle furnace and hot air oven surfaces stay hot for a long time. Use tongs and heat resistant gloves. Handle acids and solvents in a fume hood and add acid to water, never the reverse. Wear an apron and eye protection. Label every solution. Dispose of the residues of heavy metal testing as hazardous waste. Taste a sample only when the teacher directs it, and never taste an Anjana, an eye or ear preparation, or a Kshara meant for external use.

PROCEDURE

The following steps were carried out.

1. One formulation was selected from each category named in the title.
2. Each sample was received, labelled and its details were recorded.
3. The organoleptic characters of every sample were examined.
4. Each sample was weighed accurately for each test.
5. Every test was performed by the pharmacopoeial method.
6. The readings were recorded and the calculations were made.
7. The observed values were compared with the limits.
8. The inference was drawn for every parameter of every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples selected, one from each category

NO.	CATEGORY	NAME OF THE FORMULATION	MANUFACTURER	BATCH NUMBER	REFERENCE MONOGRAPH
1	Tablet				
2	Pill (Vati or Gutika)				
3	Varti				

Table 2. Organoleptic characters of every sample

SAMPLE	COLOUR	ODOUR	TASTE	TOUCH	APPEARANCE
Tablet					
Pill (Vati or Gutika)					
Varti					

Table 3. Physicochemical parameters

NO.	TEST	PHARMACOPOEIAL LIMIT	S1	S2	S3
1	Description				
2	Average weight				
3	Uniformity of weight				
4	Hardness				
5	Friability				
6	Disintegration time				
7	Loss on drying				
8	Total ash				
9	Acid insoluble ash				
10	Assay of the marker, where available				
11					
12					
13					

Table 4. Uniformity of weight

UNIT NO.	SAMPLE	WEIGHT (mg)	DEVIATION FROM THE MEAN (mg)	DEVIATION (%)	WITHIN THE LIMIT
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Table 5. Result against the pharmacopoeial limits

SAMPLE	CATEGORY	PARAMETER NOT COMPLYING, IF ANY	COMPLIES / DOES NOT COMPLY
S1	Tablet		
S2	Pill (Vati or Gutika)		
S3	Varti		

CALCULATION

Average weight = Total weight of the units ÷ Number of units

Deviation (%) = (Individual weight − Average weight) ÷ Average weight × 100

Friability (%) = (Initial weight − Final weight) ÷ Initial weight × 100

Loss on drying (%) = (Initial weight − Final weight) ÷ Initial weight × 100

Total ash (%) = Weight of the ash ÷ Weight of the sample taken × 100

Acid insoluble ash (%) = Weight of the acid insoluble ash ÷ Weight of the sample taken × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.5 — **4 hours**

Demonstration of quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana, Lauha/Mandura

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the quality control testing of one Rasoushadhi selected from each category named above, as per the parameters prescribed in the Pharmacopoeia, and to state whether each sample complies with the limits.

SESSION DETAILS

Number of samples tested			
Reference pharmacopoeia used			
Session or batch number		Date of the session	

S1 to S4 in the tables below stand for the samples listed in Table 1, in that order.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Muffle furnace and hot air oven surfaces stay hot for a long time. Use tongs and heat resistant gloves. Handle acids and solvents in a fume hood and add acid to water, never the reverse. Wear an apron and eye protection. Label every solution. Dispose of the residues of heavy metal testing as hazardous waste. Taste a sample only when the teacher directs it, and never taste an Anjana, an eye or ear preparation, or a Kshara meant for external use.

PROCEDURE

The following steps were carried out.

1. One formulation was selected from each category named in the title.
2. Each sample was received, labelled and its details were recorded.
3. The organoleptic characters of every sample were examined.
4. Each sample was weighed accurately for each test.
5. Every test was performed by the pharmacopoeial method.
6. The readings were recorded and the calculations were made.
7. The observed values were compared with the limits.
8. The inference was drawn for every parameter of every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples selected, one from each category

NO.	CATEGORY	NAME OF THE FORMULATION	MANUFACTURER	BATCH NUMBER	REFERENCE MONOGRAPH
1	Khalveeya Rasayana				
2	Parpati Rasayana				
3	Kupipakwa Rasayana				
4	Lauha or Mandura				

Table 2. Organoleptic characters of every sample

SAMPLE	COLOUR	ODOUR	TASTE	TOUCH	APPEARANCE
Khalveeya Rasayana					
Parpati Rasayana					
Kupipakwa Rasayana					
Lauha or Mandura					

Table 3. Physicochemical parameters

NO.	TEST	PHARMACOPOEIAL LIMIT	S1	S2	S3	S4
1	Loss on drying					
2	Total ash					
3	Acid insoluble ash					
4	pH, at the concentration given in the monograph					
5	Particle size					
6	Solubility					
7	Presence of free or unbound metal					
8	Assay of the metal by AAS or ICP, where available					
9						
10						
11						

Table 4. Classical tests applied to each sample

NO.	TEST	S1	S2	S3	S4	INFERENCE
1	Nischandratva					
2	Rekhapurnatva					
3	Varitaratva					
4	Unama					
5	Slakshnatva					
6	Nirdhumatva					
7	Apunarbhava					
8	Niruttha					
9	Gatarasatva					
10	Avami					
11						

Table 5. Result against the pharmacopoeial limits

SAMPLE	CATEGORY	PARAMETER NOT COMPLYING, IF ANY	COMPLIES / DOES NOT COMPLY
S1	Khalveeya Rasayana		
S2	Parpati Rasayana		
S3	Kupipakwa Rasayana		
S4	Lauha or Mandura		

CALCULATION

Loss on drying (%) = (Initial weight − Final weight) ÷ Initial weight × 100

Total ash (%) = Weight of the ash ÷ Weight of the sample taken × 100

Acid insoluble ash (%) = Weight of the acid insoluble ash ÷ Weight of the sample taken × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.6 — **2 hours**

Demonstrate Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopoeial standards

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the quality control testing of one semisolid dosage form selected from each category named above, as per the parameters prescribed in the Pharmacopoeia, and to state whether each sample complies with the limits.

SESSION DETAILS

Number of samples tested			
Reference pharmacopoeia used			
Session or batch number		Date of the session	

S1 to S4 in the tables below stand for the samples listed in Table 1, in that order.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Muffle furnace and hot air oven surfaces stay hot for a long time. Use tongs and heat resistant gloves. Handle acids and solvents in a fume hood and add acid to water, never the reverse. Wear an apron and eye protection. Label every solution. Dispose of the residues of heavy metal testing as hazardous waste. Taste a sample only when the teacher directs it, and never taste an Anjana, an eye or ear preparation, or a Kshara meant for external use.

PROCEDURE

The following steps were carried out.

1. One formulation was selected from each category named in the title.
2. Each sample was received, labelled and its details were recorded.
3. The organoleptic characters of every sample were examined.
4. Each sample was weighed accurately for each test.
5. Every test was performed by the pharmacopoeial method.
6. The readings were recorded and the calculations were made.
7. The observed values were compared with the limits.
8. The inference was drawn for every parameter of every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples selected, one from each category

NO.	CATEGORY	NAME OF THE FORMULATION	MANUFACTURER	BATCH NUMBER	REFERENCE MONOGRAPH
1	Rasakriya				
2	Avaleha				
3	Paka				
4	Anjana				

Table 2. Organoleptic characters of every sample

SAMPLE	COLOUR	ODOUR	TASTE	TOUCH	APPEARANCE
Rasakriya					
Avaleha					
Paka					
Anjana					

Table 3. Physicochemical parameters

NO.	TEST	PHARMACOPOEIAL LIMIT	S1	S2	S3	S4
1	Description					
2	pH, at the concentration given in the monograph					
3	Loss on drying					
4	Total solids					
5	Total ash					
6	Acid insoluble ash					
7	Total sugar, where applicable					
8	Reducing sugar, where applicable					
9	Specific gravity					
10	Spreadability or consistency					
11	Microbial limit					
12						
13						
14						

Table 4. Result against the pharmacopoeial limits

SAMPLE	CATEGORY	PARAMETER NOT COMPLYING, IF ANY	COMPLIES / DOES NOT COMPLY
S1	Rasakriya		
S2	Avaleha		
S3	Paka		
S4	Anjana		

CALCULATION

Loss on drying (%) = (Initial weight − Final weight) ÷ Initial weight × 100

Total ash (%) = Weight of the ash ÷ Weight of the sample taken × 100

Acid insoluble ash (%) = Weight of the acid insoluble ash ÷ Weight of the sample taken × 100

Total solids (%) = Weight of the residue ÷ Weight of the sample taken × 100

Specific gravity = Weight of the sample ÷ Weight of an equal volume of water

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.7 — **2 hours**

Demonstration of quality control parameters for liquid dosage forms like Kashaya, Arka, Sharkara, Asava-arishta

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the quality control testing of one liquid dosage form selected from each category named above, as per the parameters prescribed in the Pharmacopoeia, and to state whether each sample complies with the limits.

SESSION DETAILS

Number of samples tested			
Reference pharmacopoeia used			
Session or batch number		Date of the session	

S1 to S4 in the tables below stand for the samples listed in Table 1, in that order.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Muffle furnace and hot air oven surfaces stay hot for a long time. Use tongs and heat resistant gloves. Handle acids and solvents in a fume hood and add acid to water, never the reverse. Wear an apron and eye protection. Label every solution. Dispose of the residues of heavy metal testing as hazardous waste. Taste a sample only when the teacher directs it, and never taste an Anjana, an eye or ear preparation, or a Kshara meant for external use.

PROCEDURE

The following steps were carried out.

1. One formulation was selected from each category named in the title.
2. Each sample was received, labelled and its details were recorded.
3. The organoleptic characters of every sample were examined.
4. Each sample was weighed accurately for each test.
5. Every test was performed by the pharmacopoeial method.
6. The readings were recorded and the calculations were made.
7. The observed values were compared with the limits.
8. The inference was drawn for every parameter of every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples selected, one from each category

NO.	CATEGORY	NAME OF THE FORMULATION	MANUFACTURER	BATCH NUMBER	REFERENCE MONOGRAPH
1	Kashaya				
2	Arka				
3	Sharkara				
4	Asava-Arishta				

Table 2. Organoleptic characters of every sample

SAMPLE	COLOUR	ODOUR	TASTE	TOUCH	APPEARANCE
Kashaya					
Arka					
Sharkara					
Asava-Arishta					

Table 3. Physicochemical parameters

NO.	TEST	PHARMACOPOEIAL LIMIT	S1	S2	S3	S4
1	Description					
2	Specific gravity					
3	pH					
4	Total solids					
5	Refractive index					
6	Alcohol content, for Asava and Arishta					
7	Total sugar					
8	Reducing sugar					
9	Methanol content					
10	Total ash					
11	Microbial limit					
12						
13						
14						

Table 4. Result against the pharmacopoeial limits

SAMPLE	CATEGORY	PARAMETER NOT COMPLYING, IF ANY	COMPLIES / DOES NOT COMPLY
S1	Kashaya		
S2	Arka		
S3	Sharkara		
S4	Asava-Arishta		

CALCULATION

Total solids (%) = Weight of the residue ÷ Weight of the sample taken × 100

Specific gravity = Weight of the liquid ÷ Weight of an equal volume of water

Alcohol content (% v/v) = Volume of alcohol ÷ Total volume × 100

Total ash (%) = Weight of the ash ÷ Weight of the sample taken × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 7	Classical and Advanced quality control techniques for Ayurveda Formulations

PRACTICAL TRAINING 7.8 — **2 hours**

Demonstration of quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To carry out the quality control testing of one ophthalmic, aural or aerosol preparation selected from each category named above, as per the parameters prescribed in the Pharmacopoeia, and to state whether each sample complies with the limits.

SESSION DETAILS

Number of samples tested			
Reference pharmacopoeia used			
Session or batch number		Date of the session	

S1 to S3 in the tables below stand for the samples listed in Table 1, in that order.

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Muffle furnace and hot air oven surfaces stay hot for a long time. Use tongs and heat resistant gloves. Handle acids and solvents in a fume hood and add acid to water, never the reverse. Wear an apron and eye protection. Label every solution. Dispose of the residues of heavy metal testing as hazardous waste. Taste a sample only when the teacher directs it, and never taste an Anjana, an eye or ear preparation, or a Kshara meant for external use.

PROCEDURE

The following steps were carried out.

1. One formulation was selected from each category named in the title.
2. Each sample was received, labelled and its details were recorded.
3. The organoleptic characters of every sample were examined.
4. Each sample was weighed accurately for each test.
5. Every test was performed by the pharmacopoeial method.
6. The readings were recorded and the calculations were made.
7. The observed values were compared with the limits.
8. The inference was drawn for every parameter of every sample.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Samples selected, one from each category

NO.	CATEGORY	NAME OF THE FORMULATION	MANUFACTURER	BATCH NUMBER	REFERENCE MONOGRAPH
1	Netra Bindu				
2	Karna Bindu				
3	Aerosol or spray				

Table 2. Organoleptic characters of every sample

SAMPLE	COLOUR	ODOUR	CLARITY	APPEARANCE	CONDITION OF THE CONTAINER
Netra Bindu					
Karna Bindu					
Aerosol or spray					

Table 3. Physicochemical parameters

NO.	TEST	PHARMACOPOEIAL LIMIT	S1	S2	S3
1	Description				
2	Clarity of the solution				
3	pH				
4	Sterility test				
5	Particulate matter				
6	Tonicity				
7	Fill volume				
8	Leak test				
9	Spray pattern, for sprays				
10	Net content per actuation				
11	Pressure test, for aerosols				
12					
13					
14					

Table 4. Result against the pharmacopoeial limits

SAMPLE	CATEGORY	PARAMETER NOT COMPLYING, IF ANY	COMPLIES / DOES NOT COMPLY
S1	Netra Bindu		
S2	Karna Bindu		
S3	Aerosol or spray		

CALCULATION

Deviation of the fill volume (%) = (Volume found − Labelled volume) ÷ Labelled volume × 100

Net content per actuation = (Weight of the container before the actuations − Weight after the actuations) ÷ Number of actuations

Leak rate (mg per day) = (Initial weight of the container − Final weight of the container) ÷ Number of days stored

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 8

Nano-science, nano-technology and its application in Ayurveda

2 credits · 60 notional hours · 20 hours of practical training
5 Practical Training Activities

PRACTICAL TRAINING ACTIVITIES IN THIS MODULE

NO.	ACTIVITY	HOURS
8.1	Demonstration of the measurement of micro-particles using relevant equipment	4
8.2	Demonstration of particle size of Bhasma prepared with different numbers of Puta	4
8.3	Demonstration of particle size of various Kupipakwa Rasayana samples	4
8.4	Synthesis of silver nanoparticles (AgNPs) using the chemical precipitation method	4
8.5	Conversion of Taila into oil-in-water nanoemulsion and its characterization	4

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MODULE 8	Nano-science, nano-technology and its application in Ayurveda

PRACTICAL TRAINING 8.1

4 hours

Demonstration of the measurement of micro-particles using relevant equipment

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To measure samples of differing dimensions, from the micro to the centimetre level, using the relevant equipment, and to identify the microparticles present in them.

ALLOTMENT

Samples selected, micro to centimetre level			
Number of samples		Instruments used	
Date of the session			

REQUIREMENTS

NO.	INSTRUMENT OR MATERIAL	LEAST COUNT OR SPECIFICATION	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Vernier calipers and micrometers have sharp jaws. Close them gently. The microscope lamp becomes hot - switch it off when not in use. Handle the stage micrometer slide with care, as it is fragile and costly.

PROCEDURE

The following steps were carried out.

1. The least count of each instrument was determined.
2. The instrument was checked for a zero error.
3. Samples ranging from the micro to the centimetre level were selected.
4. Each sample was measured three times with the appropriate instrument.
5. The mean of the readings was calculated.
6. The ocular micrometer was calibrated against the stage micrometer.
7. The microparticles in the samples were identified under guidance.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Instruments used

NO.	INSTRUMENT	LEAST COUNT	RANGE	SAMPLE MEASURED WITH IT
1	Metre scale			
2	Vernier caliper			
3	Screw gauge or micrometer			
4	Stage micrometer			
5	Ocular micrometer			
6	Optical microscope			

Table 2. Measurement record

NO.	SAMPLE	INSTRUMENT USED	READING 1	READING 2	READING 3	MEAN	UNIT
1							
2							
3							
4							
5							
6							
7							
8							

Table 3. Microparticles identified

NO.	SAMPLE	STRUCTURE IDENTIFIED AS A MICROPARTICLE	SIZE (µm)	BASIS OF IDENTIFICATION
1				
2				
3				
4				
5				

CALCULATION

Least count of the vernier = 1 main scale division − 1 vernier scale division

Least count of the screw gauge = Pitch ÷ Number of divisions on the circular scale

Calibration factor of the ocular micrometer (µm per division) = (Number of stage micrometer divisions × 10) ÷ Number of ocular divisions coinciding

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 8	Nano-science, nano-technology and its application in Ayurveda

PRACTICAL TRAINING 8.2 **4 hours**

Demonstration of particle size of Bhasma prepared with different numbers of Puta

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To measure the particle size of the Bhasma samples prepared with different numbers of Puta using the relevant microscope, and to study how the particle size changes with the number of Puta given.

ALLOTMENT

Name of the Bhasma		Number of samples given	
Microscope used		Magnification used	

REQUIREMENTS

NO.	INSTRUMENT, REAGENT OR MATERIAL	SPECIFICATION	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Bhasma powder is extremely fine and becomes airborne at once. Wear a mask and handle it in a draught free area. Many Bhasma are prepared from toxic metals - do not taste any sample and wash the hands after the work. Clean the slides into a covered waste container, not into the sink.

PROCEDURE

The following steps were carried out.

1. The ocular micrometer was calibrated against the stage micrometer.
2. A small quantity of each sample was mounted on a clean slide.
3. The mount was examined under the same magnification for every sample.
4. The particle size was measured at three different fields.
5. The mean particle size was calculated for each sample.
6. The results were arranged against the number of Puta given.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Calibration of the ocular micrometer

Objective used		Stage micrometer divisions	
Ocular divisions coinciding		Calibration factor	
Unit		Date of calibration	

Table 2. Particle size measurement

NO.	SAMPLE CODE	NUMBER OF PUTA	READING 1 (µm)	READING 2 (µm)	READING 3 (µm)	MEAN (µm)	REMARK
1							
2							
3							
4							
5							
6							
7							
8							

CALCULATION

Calibration factor (µm per division) = (Number of stage micrometer divisions × 10) ÷ Number of ocular divisions coinciding

Particle size (µm) = Number of ocular divisions covered × Calibration factor

SPACE FOR THE GRAPH OF THE NUMBER OF PUTA AGAINST THE MEAN PARTICLE SIZE

TREND OBSERVED WITH THE INCREASE IN THE NUMBER OF PUTA

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 8	Nano-science, nano-technology and its application in Ayurveda

PRACTICAL TRAINING 8.3 **4 hours**

Demonstration of particle size of various Kupipakwa Rasayana samples

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To measure the particle size of the different Kupipakwa Rasayana samples using the relevant microscope, and to compare the samples with one another.

ALLOTMENT

Samples selected			
Number of samples given		Microscope used	
Magnification used			

REQUIREMENTS

NO.	INSTRUMENT, REAGENT OR MATERIAL	SPECIFICATION	QUANTITY
1			
2			
3			
4			
5			
6			
7			

SAFETY Kupipakwa Rasayana are prepared from Parada and Gandhaka and are toxic. Wear a mask and gloves and handle the powder in a draught free area. Do not taste any sample. Keep the samples in closed containers, since the fine powder escapes easily. Wash the hands after the work.

PROCEDURE

The following steps were carried out.

1. The ocular micrometer was calibrated against the stage micrometer.
2. A small quantity of each sample was mounted on a clean slide.
3. Every sample was examined under the same magnification.
4. The particle size was measured at three different fields.
5. The mean particle size was calculated for each sample.
6. The samples were compared with one another.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Calibration of the ocular micrometer

Objective used		Stage micrometer divisions	
Ocular divisions coinciding		Calibration factor	
Unit		Date of calibration	

Table 2. Particle size measurement

NO.	SAMPLE CODE	NAME OF THE KUPIPAKWA RASAYANA	READING 1 (µm)	READING 2 (µm)	READING 3 (µm)	MEAN (µm)	COLOUR AND APPEARANCE
1							
2							
3							
4							
5							
6							
7							
8							

CALCULATION

Particle size (µm) = Number of ocular divisions covered × Calibration factor

COMPARISON BETWEEN THE SAMPLES

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 8	Nano-science, nano-technology and its application in Ayurveda

PRACTICAL TRAINING 8.4 — **4 hours**

Synthesis of silver nanoparticles (AgNPs) using the chemical precipitation method

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To observe the synthesis of silver nanoparticles from a silver nitrate solution by the chemical method, and to compare the product obtained with Rajata Bhasma.

ALLOTMENT

Concentration of the silver nitrate solution		Reducing agent used	
Stabilising agent used		Volume prepared	
Method of stirring or heating		Session or batch number	

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY Silver nitrate is corrosive, stains the skin black and damages the eyes. Wear gloves, an apron and eye protection and weigh it in a fume hood. Sodium borohydride reacts with water and gives off hydrogen, which is inflammable - add it slowly and keep away from any flame. Work in a well ventilated area. Do not pour silver waste down the sink; collect it in a labelled container for recovery. Wash any splash on the skin with plenty of water at once.

PROCEDURE

The following steps were carried out.

1. All glassware was washed and rinsed with distilled water.
2. The silver nitrate solution was prepared and its concentration was recorded.
3. The reducing agent was prepared separately.
4. The reducing agent was added drop by drop with constant stirring.
5. The colour change was watched and the time was noted.
6. The stabiliser was added and stirring was continued.
7. The colloid was centrifuged, washed and stored.
8. The product was compared with a sample of Rajata Bhasma.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Reaction record

NO.	REAGENT ADDED	QUANTITY	TEMPERATURE (°C)	TIME	OBSERVATION AND COLOUR CHANGE
1					
2					
3					
4					
5					
6					
7					
8					

Table 2. Characters of the product

Colour of the colloid		Time of the colour change	
pH of the medium		Appearance after centrifugation	
Yield obtained		Storage condition	
Particle size observed			
Method used for the confirmation			

Table 3. Comparison with Rajata Bhasma

PARAMETER	SILVER NANOPARTICLES	RAJATA BHASMA
Method of preparation		
Starting material		
Colour		
Particle size		
Presence of organic matter		
Behaviour on heating		
Test applied		
Inference		

CALCULATION

Percentage of yield = Weight of the product obtained ÷ Theoretical weight × 100

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

MODULE 8	Nano-science, nano-technology and its application in Ayurveda

PRACTICAL TRAINING 8.5 — **4 hours**

Conversion of Taila into oil-in-water nanoemulsion and its characterization

Name of the student			
Roll No. / Enrolment No.			
Batch		Semester	
Date of performance		Date of submission	

AIM

To convert the selected Taila Kalpa into an oil-in-water nanoemulsion by the sonication method using a suitable surfactant, and to characterise and compare the intermediate sample and the final nanoemulsion for particle size with the help of a high-resolution microscope.

ALLOTMENT

Taila Kalpa selected		Surfactant used	
Co-surfactant used		Sonication time and amplitude	
Volume prepared		Session or batch number	

REQUIREMENTS

NO.	APPARATUS, INSTRUMENT OR REAGENT	SPECIFICATION OR GRADE	QUANTITY
1			
2			
3			
4			
5			
6			
7			
8			

SAFETY A probe sonicator produces noise above the safe limit and heats the sample. Wear ear protection and keep the vessel in an ice bath. Never run the probe in air or touch it while it is working. Some surfactants irritate the skin and the eyes - wear gloves and eye protection. Keep the sonicator inside an acoustic enclosure where one is provided.

PROCEDURE

The following steps were carried out.

1. The oil phase and the aqueous phase were prepared separately.
2. The surfactant and the co-surfactant were dissolved in the chosen phase.
3. The oil phase was added to the aqueous phase with stirring.
4. The coarse emulsion was formed and an intermediate sample was collected.
5. The emulsion was sonicated for the fixed time in an ice bath.
6. The students repeated the procedure following the teacher's instructions.
7. The final nanoemulsion was collected and stored.
8. Both samples were characterised for particle size under a high-resolution microscope and compared.

Procedure as actually carried out:

OBSERVATIONS

Table 1. Formula

NO.	INGREDIENT	FUNCTION SERVED	QUANTITY	PERCENTAGE
1				
2				
3				
4				
5				
6				
7				

Table 2. Process record

NO.	OPERATION	EQUIPMENT USED	TEMPERATURE (°C)	TIME	OBSERVATION
1					
2					
3					
4					
5					
6					
7					

Table 3. Characterisation and comparison

PARAMETER	INTERMEDIATE SAMPLE	FINAL NANOEMULSION	METHOD USED
Appearance			
Colour			
pH			
Globule size, reading 1			
Globule size, reading 2			
Globule size, mean			
Polydispersity			
Viscosity			
Phase separation after 24 hours			
Behaviour on dilution with water			

CALCULATION

Percentage of the oil phase = Volume of the oil ÷ Total volume × 100

Globule size (μm) = Number of ocular divisions covered × Calibration factor

RESULT

DISCUSSION

CONCLUSION

PRECAUTIONS

Date

Signature of the teacher

GLOSSARY

The Ayurvedic and technical terms used in these proformas. Sanskrit terms are written in plain Roman letters, without diacritical marks, as in the companion textbook.

A

Agrahya Lakshana
The features that make a raw drug unfit for use.

Amla Pariksha
A special test for Bhasma prepared from copper or from a copper-containing drug. The Bhasma is kept with a sour medium such as curd or lemon juice. A blue or green colour shows unreacted copper, and the Bhasma is then not fit for use.

Anjana
A semisolid or powdered preparation applied to the eye.

Anupana
The vehicle taken along with a medicine.

Apunarbhava
A test of Bhasma. The Bhasma, heated with certain reducing materials, does not return to the metallic state.

Arishta
A liquid preparation fermented from a decoction.

Arka
A liquid preparation obtained by distillation.

Asava
A liquid preparation fermented from a cold aqueous extract.

Avaleha
A semisolid preparation of the consistency of a linctus.

Avami
A sign of a properly prepared Bhasma, especially of Tamra Bhasma: it does not cause nausea or vomiting when taken.

B

Bharjana
Roasting or frying, used as a method of Shodhana.

Bhasma
The fine incinerated powder obtained from a metal or a mineral by Marana.

Bhavana
Wet trituration of a drug with a prescribed liquid, followed by drying.

C

Chakrika
A flattened cake made of the triturated mass, dried before the Puta.

Churna
A fine powder prepared from one or more dried drugs.

D

Dhalana
Melting a metal and pouring it into a prescribed liquid.

Dravya
A substance or a drug. A Shodhana Dravya, a Marana Dravya or a Bhavana Dravya is the material used in that process.

G

Gandhaka
Sulphur.

Gatarasatva
A test of Bhasma. The Bhasma is without taste.

Grahya Lakshana
The features that make a raw drug fit for use.

Guna
A property of a drug, such as heavy or light, hot or cold, or oily or dry.

Gutika
A pill. The term is used almost interchangeably with Vati.

H

Haratala
Orpiment, a sulphide of arsenic.

K

Kajjali
The black fine powder obtained by triturating Parada with Gandhaka.

Kalpa
A formulation, or the method by which it is prepared, as in Taila Kalpa.

Kalpana
A method of pharmaceutical preparation, and the dosage form it yields.

Karna Bindu
Drops meant for instillation into the ear.

Kashaya
A decoction.

Kashthoushadhi
A medicine prepared from plant drugs, as distinct from a Rasoushadhi.

Khalveeya Rasayana
A preparation made by trituration in a Khalva Yantra, without the use of heat.

Kshara
An alkaline preparation obtained from the ash of a drug.

Kupipakwa Rasayana
A preparation made by heating the drug in a glass bottle, the Kupi, in a sand bath.

Kwatha Churna
A coarse powder meant for the preparation of a decoction.

L

Lauha
A preparation containing iron.

Lavana
A salt preparation.

Lepa
A paste applied externally.

M

Manahshila
Realgar, a sulphide of arsenic.

Mandura
A preparation made from the slag of iron.

Marana
The process of converting a Shodhita metal or mineral into Bhasma.

Mardana
Trituration.

Musha
A crucible, made of clay and other materials, used for heating a drug.

N

Naga
Lead.

Netra Bindu
Drops meant for instillation into the eye.

Nighantu
A classical lexicon of drugs, giving their names, synonyms and properties.

Nimajjana
Immersion of a drug in a liquid, used as a method of Shodhana.

Nirdhumatva
A test of Bhasma. The Bhasma gives off no fume when heated.

Nirjaleekarana
Removal of water, that is, dehydration.

Niruttha
A test of Bhasma. The Bhasma heated with silver leaves the silver unchanged in weight.

Nirvapa
Quenching a heated drug in a prescribed liquid.

Nischandratva
A test of Bhasma. The Bhasma shows no metallic lustre.

P

Pachana
Boiling a drug in a prescribed liquid.

Paka
A preparation made by cooking, and the stage that cooking reaches.

Parada
Mercury.

Parpati
A thin flake-like preparation made by pouring the molten mass on a prepared surface.

Paryaya
A synonym by which a drug is known in the classical texts.

Patana
Sublimation or distillation.

Pishti
A preparation made by trituration with a prescribed liquid, without the use of fire.

Puta
The measured quantity of heat given to a sealed Sharava Samputa, and the arrangement by which it is given.

R

Rajata
Silver. Rajata Bhasma is the Bhasma prepared from it.

Rasa
Taste, one of the properties by which a drug is described. In Rasashastra the word also denotes mercury.

Rasadravya
A metal, a mineral or a similar material used in Rasashastra.

Rasakriya
A semisolid preparation obtained by concentrating a decoction.

Rasashastra
The branch of Ayurveda that deals with metals, minerals and their pharmaceutical processing.

Rasoushadhi
A medicine prepared with Rasadravyas, that is, with mercury, metals or minerals.

Rekhapurnatva
A test of Bhasma. The Bhasma fills the lines of the finger when rubbed between them.

S

Samskara
A process that changes the properties of a substance.

Satva
The essence extracted from a mineral or a plant.

Sharava Samputa
Two earthen saucers sealed face to face, holding the drug during the Puta.

Sharkara
A sugar-based liquid preparation.

Shodhana
The process of purifying a drug and removing its unwanted qualities.

Shodhita
A drug that has undergone Shodhana.

Siddhi Lakshana
The features that confirm the completion of a pharmaceutical process.

Slakshnatva
A test of Bhasma. The Bhasma feels smooth and fine to the touch.

Sneha
A fat or an oil, such as ghee or sesame oil, used as a medium, and a preparation made with it.

Sudha Varga
The group of calcium-rich drugs, such as conch shell, pearl oyster shell and cowrie.

Swangasheeta
Cooling by itself, without any outside help.

Swedana
Heating a drug placed in a liquid bath.

T

Taila
A medicated oil preparation.

Tamra
Copper.

U

Unama
A test of Bhasma. The Bhasma floating on water does not sink when a grain of rice is placed upon it.

V

Vanga
Tin.

Varga
A group under which drugs of like nature are classified.

Varitaratva
A test of Bhasma. The Bhasma floats on still water.

Varti
A wick or a suppository-like preparation.

Vati
A tablet or a pill made from the drug mass and dried.

Vipaka
The post-digestive effect of a drug, the taste it takes on after digestion.

Virya
The potency of a drug, described as hot or cold.

Visha Dravya
A drug classed as poisonous, which must be processed before use.

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Corrections and suggestions from teachers and scholars are welcome, and will be carried into the next edition.

Applied Basics of Rasashastra and Bhaishajya Kalpana

Forty-seven proformas, one for every Practical Training Activity listed in Table 4 of the NCISM Competency-Based Dynamic Curriculum for AYPG-AB-RB. The title of each is the curriculum's own, and the aim, the allotment, the requirements, the procedure and the observation tables are drawn from its activity details. What is left blank is only what the scholar must supply.

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- Classical tests set out in full, including Bhasma Siddhi Lakshana
- Safety blocks on every hazardous procedure, printed before the method begins
- A separator page for each of the eight modules
- A record of completion covering all forty-seven activities

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